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Digital Signal Processing.

Introduction

Processing a signal in digital domain is known as DSP.

Advantages of DSP over ASP:

- Accuracy
- fetchability
- storage

i/p $x(t)$ } Fourier series: signal should may periodic or Aperiodic
 Fourier Transform: signal must be a aperiodic.

$$\text{i/p} \rightarrow x(t)$$

$$\text{o/p} \rightarrow x(f) = x(\omega) \rightarrow \int x(t) e^{-j\omega t} dt$$

'Both i/p and o/p is continuous.

Discrete time FT: i/p signal may have finite or infinite length.

i/p is digital and o/p is continuous.

$$\text{i/p} \rightarrow x(n)$$

$$\text{o/p} \rightarrow x(e^{j\omega}) \rightarrow \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n}$$

Discrete Fourier Transform: It is applied only for that i/p signal must have finite length.

Both i/p and o/p is digital.

$$x(n) \xrightarrow{\text{DFT}} x(k)$$

$$x(k) \xrightarrow{\text{IDFT}} x(n)$$

'N' pt = denotes finite length. ^{i/p} digital sequence

'N' pt expression for both DFT and IDFT is called

'N' pt DFT pair equations.

N-pt DFT exp:

$$X(k) = \sum_{n=0}^{N-1} x(n) e^{-j \frac{2\pi}{N} nk}$$

$$; 0 \leq k \leq N-1$$

N-pt
DFT
pairs
eqns

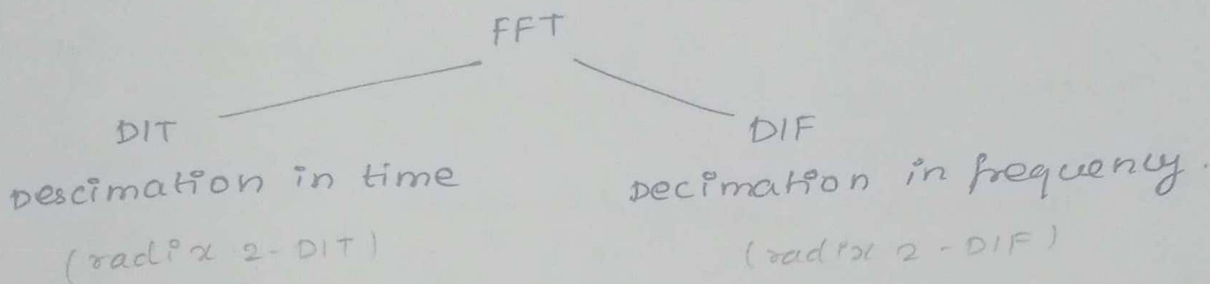
N-pt IDFT exp:

$$x(n) = \frac{1}{N} \sum_{k=0}^{N-1} X(k) e^{j \frac{2\pi}{N} nk}$$

$$; 0 \leq n \leq N-1$$

length of i/p is same as length of o/p

Fast Fourier Transform: used to compute DFT in fastest manner.



→ length of i/p = length of o/p

eg: $x(n) = \{-1, 0, 1, 1\}$, $N=4$

↓ N-pt DFT

$$X(k) = \{-1, 0, 1, 1\}$$

direct method

At $k=0 \rightarrow X(0) = x(0)e^{-j \frac{2\pi}{N} 1 \cdot 0} + x(1)e^{-j \frac{2\pi}{N} 1 \cdot 0} + x(2)e^{-j \frac{2\pi}{N} 2 \cdot 0} + x(3)e^{-j \frac{2\pi}{N} 3 \cdot 0}$

... + $x(N-1)e^{-j \frac{2\pi}{N} (N-1) \cdot 0}$ → (1)

At $k=1 \rightarrow X(1) = x(0)e^{-j \frac{2\pi}{N} 1 \cdot 1} + x(1)e^{-j \frac{2\pi}{N} 1 \cdot 1} + \dots + x(N-1)e^{-j \frac{2\pi}{N} (N-1) \cdot 1}$

At $k=N-1 \rightarrow X(N-1) = x(0)e^{-j \frac{2\pi}{N} 1 \cdot (N-1)} + x(1)e^{-j \frac{2\pi}{N} 1 \cdot (N-1)} + \dots + x(N-1)e^{-j \frac{2\pi}{N} (N-1) \cdot (N-1)}$

single point DFT
(N-1) → Addition

N → multiplication

N-pt DFT

$N(N-1) \rightarrow \text{Add} \rightarrow N(N-1)$

$N(N) \rightarrow \text{mult} \rightarrow N^2$

direct method

calculate the AM using direct method

if $N=8$

computation
comparisons

$$N! = 512$$

$$\text{Add} = N(N-1) = 8(7) = 56$$

$$\text{Add} = 512(512-1)$$

$$= 2,61,632$$

$$\text{mul} = N^2 = 64$$

$$\text{mult} = (512)^2$$

$$= 2,62,144$$

Drawback

computation comparisons

→ It requires more no. of hardware products

for addition & multiplication when length ↑

→ when length ↑ complexity ↑

⇒ Therefore speed decreases, hence also

efficiency decreases

→ Finally, Direct method is not suitable for large no. of length, then we go FFT algorithm

FFT algorithm [Fast Fourier Transform]

Matrix method (or) Linear Transformation method

$$\begin{matrix} \text{o/p matrix } x(k) & & & & & & & & & & \text{i/p matrix} \\ & & & & & & & & & & \\ & & & & & & & & & & \\ \begin{bmatrix} x(0) \\ x(1) \\ x(2) \\ \vdots \\ x(N-1) \end{bmatrix} & = & \begin{bmatrix} 1 & 1 & 1 & \dots & 1 \\ 1 & e^{-j\frac{2\pi}{N}} & \dots & \dots & e^{-j\frac{2\pi}{N}(N-1)} \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 1 & \vdots & \vdots & \vdots & \vdots \end{bmatrix} & \begin{bmatrix} x(0) \\ x(1) \\ x(2) \\ \vdots \\ x(N-1) \end{bmatrix} \\ N \times 1 & & & & & & & & & & N \times 1 \end{matrix}$$

Q) what are twiddle factors and its properties.

Twiddle factor → one of the exponential term

replace the exponential term in matrix in

terms of twiddle factor

$W_N \rightarrow$ Twiddle factor

$$W_N = e^{-j\frac{2\pi}{N}}$$

$$W = e^{-j2\pi}$$

sense to find the twiddle factor

Properties of Twiddle factor, general W_N^x ($x=0$ to $\frac{N}{2}-1$)

Periodicity = $W_N^{N+K} = W_N^K \cdot W_N^N$ if $x > N$

Symmetry = $W_N^{K+N/2} = -W_N^K \cdot W_N^{N/2}$ if $x < N$

Proof: (periodicity)

$$W_N^{K+N} = W_N^K$$

$$W_N^K \cdot W_N^N = W_N^K \left(e^{-j\frac{2\pi}{N}} \right)^N$$

$$= W_N^K (e^{-j2\pi})$$

$$= W_N^K [\underbrace{\cos 2\pi}_{(1)} - j \underbrace{\sin 2\pi}_{(0)}]$$

$$W_N^{K+N} = W_N^K$$

eg: $N=4$

$$W_4^5 \rightarrow W_4^{1+4} = W_4^1 \cdot W_4^4$$

$$e^{jn\pi} = (-1)^n$$

Proof (symmetry)

$$W_N^{K+N/2} = W_N^K \cdot W_N^{N/2}$$

$$= W_N^K \cdot W_N^{N/2} \left(e^{-j\frac{2\pi}{N}} \right)^{N/2}$$

$$= W_N^K (e^{-j\pi})$$

$$= W_N^K (\underbrace{\cos \pi}_{(-1)} - j \underbrace{\sin \pi}_0)$$

$$W_N^{K+N/2} = -W_N^K$$

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compute 'N' pt DFT for the following discrete sequences using direct method and matrix method.

1) $x(n) = \{-1, 1, 1, -1\}$ 2) $x(n) = \cos\left(\frac{n\pi}{2}\right), 0 \leq n \leq 3.$

1) $x(n) = \{-1, 1, 1, -1\}$

$N = 4$

Direct method.

$$X(k) = \sum_{n=0}^{N-1} x(n) e^{-j\frac{2\pi}{N}nk}, \quad 0 \leq k \leq N-1$$

$$X(k) = \sum_{n=0}^3 x(n) e^{-j\frac{2\pi}{4}nk}, \quad k = 0 \text{ to } 3.$$

$$X(0) = x(0) e^{-j\frac{2\pi}{4}0 \cdot 0} + x(1) e^{-j\frac{2\pi}{4}1 \cdot 0} + x(2) e^{-j\frac{2\pi}{4}2 \cdot 0} + x(3) e^{-j\frac{2\pi}{4}3 \cdot 0}$$

$$X(1) = x(0) e^{-j\frac{2\pi}{4}0 \cdot 1} + x(1) e^{-j\frac{2\pi}{4}1 \cdot 1} + x(2) e^{-j\frac{2\pi}{4}2 \cdot 1} + x(3) e^{-j\frac{2\pi}{4}3 \cdot 1}$$

$$X(2) = x(0) e^{-j\frac{2\pi}{4}0 \cdot 2} + x(1) e^{-j\frac{2\pi}{4}1 \cdot 2} + x(2) e^{-j\frac{2\pi}{4}2 \cdot 2} + x(3) e^{-j\frac{2\pi}{4}3 \cdot 2}$$

$$X(3) = x(0) e^{-j\frac{2\pi}{4}0 \cdot 3} + x(1) e^{-j\frac{2\pi}{4}1 \cdot 3} + x(2) e^{-j\frac{2\pi}{4}2 \cdot 3} + x(3) e^{-j\frac{2\pi}{4}3 \cdot 3}$$

$x(n) = \{-1, 1, 1, -1\}; \quad x(0) = -1, x(1) = 1, x(2) = 1, x(3) = -1$

If in question ask compute $N=4$, but $x(n) = \{-1, 1, 1, -1\}$

$\{-1, 1, 1, 0\}$

$\{-1, 1, 1, -1\}$

(periodic sig)

$$x(0) = x(0)e^{-0} + x(1)e^{-0} + x(2)e^{-0} + x(3)e^{-0}$$

$$x(0) = x(0) + x(1) + x(2) + x(3)$$

$$x(0) = -1 + 1 + 1 - 1$$

$$x(0) = 0 \rightarrow \textcircled{1}$$

To find $x(k)$ at $k=0$, just add all 4p points for any sequence

$$x(1) = x(0)e^{-0} + x(1)e^{-\frac{j\pi}{2}} + x(2)e^{-j\pi} + x(3)e^{-\frac{j3\pi}{2}}$$

$$e^{\pm j\theta} = \cos\theta \pm j\sin\theta$$

$$e^{-j\pi/2} = \cos\frac{\pi}{2} - j\sin\frac{\pi}{2}$$

$$= 0 - j = -j$$

$$e^{-j\pi} = \cos\pi - j\sin\pi = -1$$

$$e^{j3\pi/2} = \cos\frac{3\pi}{2} + j\sin\frac{3\pi}{2}$$

$$= 0 + (-j)$$

$$= -j$$

$$x(1) = (-1) + 1(-j) + 1(-1) + (-1)(-j)$$

$$= -1 - j - 1 + j$$

$$x(1) = -2 \rightarrow \textcircled{2}$$

$$x(2) = x(0)e^{-j\pi} + x(1)e^{-j2\pi} + x(2)e^{-j3\pi} + x(3)e^{-j4\pi}$$

$$e^{\pm jn\pi} = (-1)^n$$

$$= (-1) + 1(-1) + 1(1) + (-1)(1)$$

$$= -1 - 1 + 1 - 1$$

$$x(2) = 0 \rightarrow \textcircled{3}$$

$$x(3) = x(0)e^{-0} + x(1)e^{-\frac{j\pi 3}{2}} + x(2)e^{-j3\pi} + x(3)e^{-\frac{j9\pi}{2}}$$

$$= (-1) + 1(j) + 1(-1) + (-1)(-j)$$

$$= -1 + j - 1 + j = 2j - 2 \rightarrow \textcircled{4}$$

$$e^{-\frac{j9\pi}{2}} = \cos \frac{9\pi}{2} - j \sin \frac{9\pi}{2}$$

$$= 0 - j(1)$$

$$e^{-j9\pi/2} = -j$$

$$X(k) = \{0, -2-2j, 0, -2+2j\}$$

conjugate

In $x(k)$ exp when complex form is include then its conjugate will present

Linear transformation or matrix method

for a given

$$W_N = e^{-j2\pi/N}$$

$$X(0) = x(0)W_4^{0 \times 0} + x(1)W_4^{1 \times 0} + x(2)W_4^{2 \times 0} + x(3)W_4^{3 \times 0}$$

$$X(1) = x(0)W_4^{0 \times 1} + x(1)W_4^{1 \times 1} + x(2)W_4^{2 \times 1} + x(3)W_4^{3 \times 1}$$

$$X(2) = x(0)W_4^{0 \times 2} + x(1)W_4^{1 \times 2} + x(2)W_4^{2 \times 2} + x(3)W_4^{3 \times 2}$$

$$X(3) = x(0)W_4^{0 \times 3} + x(1)W_4^{1 \times 3} + x(2)W_4^{2 \times 3} + x(3)W_4^{3 \times 3}$$

$$\begin{bmatrix} x(0) \\ x(1) \\ x(2) \\ x(3) \end{bmatrix} = \begin{bmatrix} w_4^{0 \cdot 0} & w_4^{1 \cdot 0} & w_4^{2 \cdot 0} & w_4^{3 \cdot 0} \\ w_4^{0 \cdot 1} & w_4^{1 \cdot 1} & w_4^{2 \cdot 1} & w_4^{3 \cdot 1} \\ w_4^{0 \cdot 2} & w_4^{1 \cdot 2} & w_4^{2 \cdot 2} & w_4^{3 \cdot 2} \\ w_4^{0 \cdot 3} & w_4^{1 \cdot 3} & w_4^{2 \cdot 3} & w_4^{3 \cdot 3} \end{bmatrix} \begin{bmatrix} x(0) \\ x(1) \\ x(2) \\ x(3) \end{bmatrix}$$

$$\begin{bmatrix} x(0) \\ x(1) \\ x(2) \\ x(3) \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & w_4^{1 \cdot 1} & w_4^{2 \cdot 1} & w_4^{3 \cdot 1} \\ 1 & w_4^{1 \cdot 2} & w_4^{2 \cdot 2} & w_4^{3 \cdot 2} \\ 1 & w_4^{1 \cdot 3} & w_4^{2 \cdot 3} & w_4^{3 \cdot 3} \end{bmatrix} \begin{bmatrix} x(0) \\ x(1) \\ x(2) \\ x(3) \end{bmatrix}$$

Find $w_N^k \rightarrow w_N^{N-k}$

$$w_4^{1 \cdot 1} = e^{-\frac{j2\pi}{4} \cdot 1} = e^{-j\pi/2} = -j$$

$$w_4^{2 \cdot 2} = e^{-\frac{j2\pi}{4} \cdot 2} = e^{-j\pi} = -1$$

$$w_4^{3 \cdot 3} = e^{-\frac{j2\pi}{4} \cdot 3} = e^{-j3\pi/2} = +j$$

$$w_N^N = 1 \Rightarrow w_4^{2 \cdot 2} = w_4^{3 \cdot 3} = 1$$

$$w_4^6 = w_4^{4+2} = w_4^4 \cdot w_4^2$$

also power values is more than N, periodicity property

$$= 1 \cdot (-1) = -1$$

$$w_4^9 = w_4^{5+4} = w_4^5 \rightarrow w_4^{4+1} \rightarrow w_4^1 = -j$$

$$\begin{bmatrix} x(0) \\ x(1) \\ x(2) \\ x(3) \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \begin{bmatrix} x(0) \\ x(1) \\ x(2) \\ x(3) \end{bmatrix}$$

only for 4×4 matrix

→ First row column is one

→ Third row alternate -1 and 1

→ Fourth row is complex conjugate of second row.

$$\begin{bmatrix} x(0) \\ x(1) \\ x(2) \\ x(3) \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \begin{bmatrix} -1 \\ 1 \\ 1 \\ -1 \end{bmatrix}$$

$$\begin{bmatrix} x(0) \\ x(1) \\ x(2) \\ x(3) \end{bmatrix} = \begin{bmatrix} -1+1+1-1 \\ -1-j-1-j \\ -1-1+1+1 \\ -1+j-1+j \end{bmatrix} = \begin{bmatrix} 0 \\ -2-2j \\ 0 \\ -2+2j \end{bmatrix}$$

$$\therefore x(k) = \{ 0, -2-2j, 0, -2+2j \}$$

2. $x(n) = \cos\left(\frac{n\pi}{2}\right), 0 \leq n \leq 3$

$$x(0) = 1, x(1) = 0, x(2) = -1, x(3) = 0$$

$$x(n) = \{ 1, 0, -1, 0 \}$$

Direct method:

Same as before sum

$$x(0) = x(0)e^{-0} + x(1)e^{-0} + x(2)e^{-0} + x(3)e^{-0}$$

$$x(0) = 1 + 0 - 1 + 0 = 0$$

$$x(1) = x(0)e^{-0} + x(1)e^{-j\pi/2} + x(2)e^{-j\pi} + x(3)e^{-j3\pi/2}$$

$$= 1 + 0 + (-1)(-1)$$

$$= 2$$

$$x(2) = x(0)e^{-0} + x(1)e^{-j\pi} + x(2)e^{-j2\pi} + x(3)e^{-j3\pi}$$

$$= 1 + 0 + (-1)(1) = 0$$

$$x(3) = x(0)e^{-0} + x(1)e^{-j\pi/2} + x(2)e^{-j3\pi/2} + x(3)e^{-j9\pi/2}$$

$$= 1 + 0 + (-1)(-1)$$

$$= 2$$

$$x(k) = \{0, 2, 0, 2\}$$

Matrix method:

$$\begin{bmatrix} x(0) \\ x(1) \\ x(2) \\ x(3) \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & W_4 = -j & W_4^2 = -1 & j \\ 1 & 1 & -1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \begin{bmatrix} x(0) \\ x(1) \\ x(2) \\ x(3) \end{bmatrix}$$

$$\begin{bmatrix} x(0) \\ x(1) \\ x(2) \\ x(3) \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1+0-1+0 \\ 1+0+1+0 \\ 1+0-1+0 \\ 1+0+1+0 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \\ 0 \\ 2 \end{bmatrix}$$

$$x(k) = \{0, 2, 0, 2\}$$

$$x(n) = \{4, -2, 2, 2\}$$

$$x(0) = x(0)e^{-0} + x(1)e^{-0} + x(2)e^{-0} + x(3)e^{-0}$$

$$= 4 - 2 + 2 + 2 = 6$$

$$x(1) = x(0)e^{-0} + x(1)e^{-j\pi/2} + x(2)e^{-j\pi} + x(3)e^{-j3\pi/2}$$

$$= 4 + (-2)(-j) + (2)(-1) + 2(j)$$

$$= 4 + 2j - 2 + 2j \rightarrow 2 + 4j$$

$$x(2) = x(0)e^{-0} + x(1)e^{-j\pi} + x(2)e^{-j2\pi} + x(3)e^{-j3\pi}$$

$$= 4 + (-2)(-1) + 2(1) + 2(-1)$$

$$= 4 + 2 + 2 - 2$$

$$x(2) = 6$$

$$x(3) = x(0)e^{-0} + x(1)e^{-j3\pi/2} + x(2)e^{-j3\pi} + x(3)e^{-j9\pi/2}$$

$$= 4 + (-2)(j) + 2(-1) + 2(j)$$

$$= 4 - 2j - 2 - 2j$$

$$x(3) = 2 - 4j$$

$$\therefore x(k) = \{6, 2+4j, 6, 2-4j\}$$

Matrix method (4x4) same as before sum

$$\begin{bmatrix} x(1) \\ x(2) \\ x(3) \\ x(4) \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \begin{bmatrix} 4 \\ -2 \\ 2 \\ 2 \end{bmatrix}$$

$$= \begin{bmatrix} 4 - 2 + 2 + 2 \\ 4 + 2j - 2 + 2j \\ 4 + 2 + 2 - 2 \\ 4 - 2j - 2 - 2j \end{bmatrix} = \begin{bmatrix} 6 \\ 2 + 4j \\ 6 \\ 2 - 4j \end{bmatrix}$$

$$\therefore x(k) = \{6, 2+4j, 6, 2-4j\}$$

4. $x(n) = (-1)^n$; $0 \leq n \leq 3$ (0 to 3)

$$x(n) = \{1, -1, 1, -1\}$$

$$x(0) = x(0)e^{-0} + x(1)e^{-0} + x(2)e^{-0} + x(3)e^{-0}$$

$$= 1 - 1 + 1 - 1 = 0$$

$$\begin{aligned}
 x(1) &= x(0)e^{-0} + x(1)e^{-j\pi/2} + x(2)e^{-j\pi} + x(3)e^{-j3\pi/2} \\
 &= 1 + (-1)(-j) + 1(-1) + (-1)(j) \\
 &= 1 + j - 1 - j \\
 &= 0
 \end{aligned}$$

$$\begin{aligned}
 x(2) &= x(0)e^{-0} + x(1)e^{-j\pi} + x(2)e^{-j2\pi} + x(3)e^{-j3\pi} \\
 &= 1 + (-1)(-1) + 1(1) + (-1)(-1) \\
 &= 1 + 1 + 1 + 1 \\
 &= 4
 \end{aligned}$$

$$\begin{aligned}
 x(3) &= x(0)e^{-0} + x(1)e^{-j3\pi/2} + x(2)e^{-j3\pi} + x(3)e^{-j9\pi/2} \\
 &= 1 + (-1)(-j) + 1(-1) + (-1)(j) \\
 &= 1 - j - 1 + j \\
 &= 0
 \end{aligned}$$

$$x(3) = 0$$

$$x(k) = \{0, 0, 4, 0\}$$

Matrix method

$$\begin{bmatrix} x(1) \\ x(2) \\ x(3) \\ x(4) \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ -1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix}$$

$$= \begin{bmatrix} 1-1+1-1 \\ 1+j-1-j \\ 1-1+1-1 \\ 1-j-1+j \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$x(k) = \{0, 0, 4, 0\}$$

'N' pt IDFT.

$$x(n) = \frac{1}{N} \sum_{k=0}^{N-1} x(k) e^{\frac{j2\pi}{N} nk}$$

$n=0$ to $N-1$

$k=0$ to $N-1$

$$W_N = \frac{e^{-j2\pi}}{N}$$

$$x(n) = \frac{1}{N} \sum_{k=0}^{N-1} x(k) W_N^{-nk}$$

$n=0$ to $N-1$

if $N=4$

4 pt DFT

$$\begin{bmatrix} x(0) \\ x(1) \\ x(2) \\ x(3) \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \begin{bmatrix} x(0) \\ x(1) \\ x(2) \\ x(3) \end{bmatrix}$$

4x4

4 pt IDFT. [twiddle factors in both DFT & IDFT is complex conjugate]

$$\begin{bmatrix} x(0) \\ x(1) \\ x(2) \\ x(3) \end{bmatrix} = \frac{1}{N} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & j & -1 & -j \\ 1 & -1 & 1 & -1 \\ 1 & -j & -1 & j \end{bmatrix} \begin{bmatrix} x(0) \\ x(1) \\ x(2) \\ x(3) \end{bmatrix}$$

4x4

1. Compute N'pt IDFT the following discrete spectrum using linear transformation method.

(ii) $x(k) = \{0, -2-j2, 0, -2+j2\}$

$$x(n) = \frac{1}{N} \sum_{k=0}^{N-1} X(k) W_N^{-nk}$$

$$\begin{bmatrix} x(0) \\ x(1) \\ x(2) \\ x(3) \end{bmatrix} = \frac{1}{4} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & j & -1 & -j \\ 1 & -1 & 1 & -1 \\ 1 & -j & 1 & j \end{bmatrix} \begin{bmatrix} 0 \\ -2-j2 \\ 0 \\ -2+j2 \end{bmatrix}$$

$$= \frac{1}{4} \begin{bmatrix} 0 - 2 - j2 + 0 - 2 + j2 \\ 0 - 2j + 2 + 0 + 2j + 2 \\ 0 + 2 + j2 + 0 + 2 - j2 \\ 0 + 2j - 2 + 0 - 2j - 2 \end{bmatrix} = \frac{1}{4} \begin{bmatrix} -4 \\ 4 \\ 4 \\ -4 \end{bmatrix}$$

$$\begin{bmatrix} x(0) \\ x(1) \\ x(2) \\ x(3) \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ 1 \\ -1 \end{bmatrix}$$

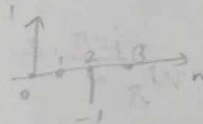
sequence repre

$$x(n) = \{-1, 1, 1, -1\}$$

functional repre

$$x(n) = \cos(n\pi/2)$$

graphical rep



IDFT

$$X(k) = \{0, 2, 0, 2\}$$

$$\begin{bmatrix} x(0) \\ x(1) \\ x(2) \\ x(3) \end{bmatrix} = \frac{1}{4} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & j & -1 & -j \\ 1 & -1 & 1 & -1 \\ 1 & -j & 1 & j \end{bmatrix} \begin{bmatrix} 0 \\ 2 \\ 0 \\ 2 \end{bmatrix} = \frac{1}{4} \begin{bmatrix} 0 + 2 + 0 + 2 \\ 0 + 2j + 0 - 2j \\ 0 - 2 + 0 - 2 \\ 0 - 2j + 0 + 2j \end{bmatrix}$$

$$\begin{bmatrix} x(0) \\ x(1) \\ x(2) \\ x(3) \end{bmatrix} = \frac{1}{4} \begin{bmatrix} 4 \\ 0 \\ -4 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ -1 \\ 0 \end{bmatrix}$$

Construct twiddle factor matrix for 8pt DFT
and 8 IDFT

8 pt DFT

$$\begin{bmatrix} x(0) \\ x(1) \\ x(2) \\ x(3) \\ x(4) \\ x(5) \\ x(6) \\ x(7) \end{bmatrix} = \begin{bmatrix} 1.0 & 1.0 & 1.0 & 1.0 & 1.0 & 1.0 & 1.0 & 1.0 \\ 0.0 & 1.0 & 2.0 & 3.0 & 4.0 & 5.0 & 6.0 & 7.0 \\ 0.1 & 1.1 & 2.1 & 3.1 & 4.1 & 5.1 & 6.1 & 7.1 \\ 0.2 & 1.2 & 2.2 & 3.2 & 4.2 & 5.2 & 6.2 & 7.2 \\ 0.3 & 1.3 & 2.3 & 3.3 & 4.3 & 5.3 & 6.3 & 7.3 \\ 0.4 & 1.4 & 2.4 & 3.4 & 4.4 & 5.4 & 6.4 & 7.4 \\ 0.5 & 1.5 & 2.5 & 3.5 & 4.5 & 5.5 & 6.5 & 7.5 \\ 0.6 & 1.6 & 2.6 & 3.6 & 4.6 & 5.6 & 6.6 & 7.6 \\ 0.7 & 1.7 & 2.7 & 3.7 & 4.7 & 5.7 & 6.7 & 7.7 \end{bmatrix} \begin{bmatrix} W_8^0 \\ W_8^1 \\ W_8^2 \\ W_8^3 \\ W_8^4 \\ W_8^5 \\ W_8^6 \\ W_8^7 \end{bmatrix}$$

To find $W_8^1, W_8^2, W_8^3, W_8^4$

$$W_8^0 = 1$$

$$W_8^1 = e^{-\frac{j2\pi}{8}} = e^{-j\pi/4} = \cos \pi/4 - j \sin \pi/4 = 0.707 - j0.707 = \frac{1}{\sqrt{2}} - j\frac{1}{\sqrt{2}}$$

$$W_8^2 = W_8^{-\frac{j2\pi}{8} \cdot 2} = W_8^{-j\pi/2} = -j$$

$$\begin{aligned}
 W_8^3 &= W_8^3 = W_8^{2+1} = W_8^2 \cdot W_8^1 \\
 &= (-j)(0.707 - j0.707) \\
 &= -0.707j + 0.707 \\
 &= 0.707(1 - j)
 \end{aligned}$$

$$\begin{aligned}
 W_8^4 &= e^{-\frac{j2\pi}{8} \times 4} = e^{-j\pi} = \cos \pi - j \sin \pi \\
 &= -1
 \end{aligned}$$

$$\begin{aligned}
 W_8^5 &= W_8^{4+1} = W_8^4 \cdot W_8^1 = -1 \cdot (0.707 - j0.707) = -0.707 + j0.707 \\
 &= 0.707(-1 + j)
 \end{aligned}$$

$$W_8^5 = e^{-j\frac{2\pi}{8} \times 5} = e^{-j5\pi/4} = \cos \frac{5\pi}{4} - j \sin \frac{5\pi}{4}$$

$$W_8^5 = W_8^{15/2} = -W_8^3 = -\frac{1}{\sqrt{2}} + \frac{j}{\sqrt{2}}$$

$$W_8^6 = e^{-j\frac{2\pi}{8} \times 6} = e^{-j3\pi/2} = \cos \frac{3\pi}{2} - j \sin \frac{3\pi}{2}$$

$$= 0 - j(-1)$$

$$= j$$

$$W_8^7 = e^{-j\frac{2\pi}{8} \times 7} = e^{-j7\pi/4} = \cos \frac{7\pi}{4} - j \sin \frac{7\pi}{4}$$

$$= \frac{1}{\sqrt{2}} + \frac{j}{\sqrt{2}}$$

$$W_8^8 = 1$$

$$W_8^9 = W_8^{8+1} = W_8^8 \cdot W_8^1 = 1 \cdot \left(\frac{1}{\sqrt{2}} - j\frac{1}{\sqrt{2}}\right) = \frac{1}{\sqrt{2}} - j\frac{1}{\sqrt{2}}$$

$$W_8^{10} = W_8^{8+2} = W_8^8 \cdot W_8^2 = 1 \cdot (-j) = -j$$

$$W_8^{12} = W_8^{8+4} = W_8^8 \cdot W_8^4 = 1 \cdot (-1) = -1$$

$$W_8^{14} = W_8^{8+6} = W_8^8 \cdot W_8^6 = 1 \cdot (j) = j$$

$$W_8^{15} = W_8^{8+7} = W_8^8 \cdot W_8^7 = 1 \cdot \left[\frac{1}{\sqrt{2}} + \frac{j}{\sqrt{2}}\right] = \frac{1}{\sqrt{2}} + \frac{j}{\sqrt{2}}$$

$$W_8^{16} = W_8^{8+8} = W_8^8 \cdot W_8^8 = 1$$

$$W_8^{18} = W_8^{8+10} = W_8^8 \cdot W_8^{10} = 1 \cdot (-j) = -j$$

$$W_8^{20} = W_8^{8+12} = W_8^8 \cdot W_8^{12} = 1 \cdot (-1) = -1$$

$$W_8^{21} = W_8^{8+13} = W_8^8 \cdot W_8^{13} = 1 \cdot \left(-\frac{1}{\sqrt{2}} + \frac{j}{\sqrt{2}}\right) = -\frac{1}{\sqrt{2}} + \frac{j}{\sqrt{2}}$$

$$W_8^{24} = W_8^{8+16} = W_8^8 \cdot W_8^{16} = 1$$

$$W_8^{25} = W_8^{8+17} = W_8^8 \cdot W_8^{17} = 1 \cdot \left(\frac{1}{\sqrt{2}} - j\frac{1}{\sqrt{2}}\right) = \frac{1}{\sqrt{2}} - j\frac{1}{\sqrt{2}}$$

$$W_8^{28} = W_8^{8+20} = W_8^8 \cdot W_8^{20} = 1 \cdot (-1) = -1$$

$$W_8^{30} = W_8^{8+22} = W_8^8 \cdot W_8^{22} = 1 \cdot j = j$$

$$W_8^0 \cdot W_8^8 \cdot W_8^8 \cdot W_8^8 = W_8^0 = 1$$

$$W_8^{35} = W_8^{8+27} = W_8^8 \cdot W_8^{27} = 1 \cdot W_8^{19} = W_8^{8+11} = W_8^8 = \frac{1}{\sqrt{2}} - \frac{j}{\sqrt{2}}$$

$$W_8^{36} = W_8^{8+28} = 1 \cdot (-1) = -1$$

$$W_8^{42} = W_8^{8+34} = 1 \cdot W_8^{26} = 1 \cdot W_8^{18} = -j$$

$$W_8^{49} = W_8^{8+41} = 1 \cdot W_8^{33} = W_8^{8+25} = \frac{1}{\sqrt{2}} - j \frac{1}{\sqrt{2}}$$

8-pt DFT

$$\begin{bmatrix} X(0) \\ X(1) \\ X(2) \\ X(3) \\ X(4) \\ X(5) \\ X(6) \\ X(7) \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ & \frac{1}{\sqrt{2}} - \frac{j}{\sqrt{2}} & -j & -\frac{1}{\sqrt{2}} - \frac{j}{\sqrt{2}} & -1 & -\frac{1}{\sqrt{2}} + \frac{j}{\sqrt{2}} & j & \frac{1}{\sqrt{2}} + \frac{j}{\sqrt{2}} \\ & & -j & -1 & j & 1 & -j & -1 \\ & & & \frac{1}{\sqrt{2}} - \frac{j}{\sqrt{2}} & j & \frac{1}{\sqrt{2}} - \frac{j}{\sqrt{2}} & -1 & \frac{1}{\sqrt{2}} + \frac{j}{\sqrt{2}} \\ & & & & -1 & 1 & -1 & 1 \\ & & & & & \frac{1}{\sqrt{2}} + \frac{j}{\sqrt{2}} & -j & \frac{1}{\sqrt{2}} - \frac{j}{\sqrt{2}} \\ & & & & & & j & -\frac{1}{\sqrt{2}} - \frac{j}{\sqrt{2}} \\ & & & & & & & \frac{1}{\sqrt{2}} - \frac{j}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} x(0) \\ x(1) \\ x(2) \\ x(3) \\ x(4) \\ x(5) \\ x(6) \\ x(7) \end{bmatrix}$$

8-pt IDFT

$$\begin{bmatrix} x(0) \\ x(1) \\ x(2) \\ x(3) \\ x(4) \\ x(5) \\ x(6) \\ x(7) \end{bmatrix} = \frac{1}{N} \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ & \frac{1}{\sqrt{2}} + \frac{j}{\sqrt{2}} & j & \frac{1}{\sqrt{2}} + \frac{j}{\sqrt{2}} & -1 & \frac{1}{\sqrt{2}} - \frac{j}{\sqrt{2}} & -j & \frac{1}{\sqrt{2}} - \frac{j}{\sqrt{2}} \\ & & j & -1 & -j & 1 & j & -1 \\ & & & \frac{1}{\sqrt{2}} + \frac{j}{\sqrt{2}} & -j & \frac{1}{\sqrt{2}} + \frac{j}{\sqrt{2}} & -1 & \frac{1}{\sqrt{2}} - \frac{j}{\sqrt{2}} \\ & & & & -1 & 1 & -1 & 1 \\ & & & & & \frac{1}{\sqrt{2}} - \frac{j}{\sqrt{2}} & j & \frac{1}{\sqrt{2}} + \frac{j}{\sqrt{2}} \\ & & & & & & j & -\frac{1}{\sqrt{2}} - \frac{j}{\sqrt{2}} \\ & & & & & & & \frac{1}{\sqrt{2}} + \frac{j}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} X(0) \\ X(1) \\ X(2) \\ X(3) \\ X(4) \\ X(5) \\ X(6) \\ X(7) \end{bmatrix}$$

find DFT for $x(n) = \{0, 1, 2, 3, 4, 5, 6, 7\}$; $0 \leq n \leq 7$

soln matrix method $N=8$

$$\begin{bmatrix} x(0) \\ x(1) \\ x(2) \\ x(3) \\ x(4) \\ x(5) \\ x(6) \\ x(7) \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & \frac{1-j}{\sqrt{2}} & -j & \frac{-1-j}{\sqrt{2}} & -1 & \frac{-1+j}{\sqrt{2}} & j & \frac{1+j}{\sqrt{2}} \\ 1 & -j & -1 & j & 1 & -j & -1 & j \\ 1 & \frac{-1-j}{\sqrt{2}} & j & \frac{1-j}{\sqrt{2}} & -1 & \frac{1+j}{\sqrt{2}} & -j & \frac{-1+j}{\sqrt{2}} \\ 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 \\ 1 & \frac{-1+j}{\sqrt{2}} & -j & \frac{1+j}{\sqrt{2}} & -1 & \frac{-j}{\sqrt{2}} & j & \frac{-1-j}{\sqrt{2}} \\ 1 & j & -1 & -j & 1 & j & -1 & -j \\ 1 & \frac{1+j}{\sqrt{2}} & j & \frac{-1+j}{\sqrt{2}} & -1 & \frac{-1-j}{\sqrt{2}} & -j & \frac{1-j}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \\ 7 \end{bmatrix}$$

$$= \begin{bmatrix} 0+1+2+3+4+5+6+7 \\ 0 + \frac{1-j}{\sqrt{2}} - 2j - \frac{3-j}{\sqrt{2}} - 4 - \frac{5}{\sqrt{2}} + \frac{5j}{\sqrt{2}} + 6j + \frac{7}{\sqrt{2}} + \frac{7j}{\sqrt{2}} \\ 0 - j - 2 + 3j + 4 - 5j - 6 + 7j \\ 0 - \frac{1}{\sqrt{2}} - \frac{j}{\sqrt{2}} + 2j + \frac{3}{\sqrt{2}} - \frac{3j}{\sqrt{2}} - 4 + \frac{5}{\sqrt{2}} + \frac{5j}{\sqrt{2}} - 6j - \frac{7}{\sqrt{2}} + \frac{7j}{\sqrt{2}} \\ 0 - 1 + 2 - 3 + 4 - 5 + 6 - 7 \\ 0 - \frac{1}{\sqrt{2}} + \frac{j}{\sqrt{2}} - 2j + \frac{3}{\sqrt{2}} + \frac{3j}{\sqrt{2}} - 4 + \frac{5}{\sqrt{2}} - \frac{5j}{\sqrt{2}} + 6j - \frac{7}{\sqrt{2}} - \frac{7j}{\sqrt{2}} \\ 0 + j - 2 - 3j + 4 + 5j - 6 - 7j \\ 0 + \frac{1}{\sqrt{2}} + \frac{j}{\sqrt{2}} + 2j - \frac{3}{\sqrt{2}} + \frac{3j}{\sqrt{2}} - 4 - \frac{5}{\sqrt{2}} - \frac{5j}{\sqrt{2}} - 6j + \frac{7}{\sqrt{2}} - \frac{7j}{\sqrt{2}} \end{bmatrix}$$

$$\begin{bmatrix} x(0) \\ x(1) \\ x(2) \\ x(3) \\ x(4) \\ x(5) \\ x(6) \end{bmatrix} = \begin{bmatrix} 28 \\ \frac{8j}{\sqrt{2}} + 4j - 4 \\ 4j - 4 \\ \frac{8j}{\sqrt{2}} - 4j - 4 \\ -4 \\ -\frac{8j}{\sqrt{2}} + 4j - 4 \\ -4 - 4j \\ -\frac{8j}{\sqrt{2}} - 4j - 4 \end{bmatrix} = \begin{bmatrix} 28 \\ -4 + j9.657 \\ -4 + 4j \\ -4 + j1.657 \\ -4 \\ -4 - j1.657 \\ -4 - 4j \\ -4 - j9.657 \end{bmatrix}$$

IDFT

$$\begin{bmatrix} x(0) \\ x(1) \\ x(2) \\ x(3) \\ x(4) \\ x(5) \\ x(6) \\ x(7) \end{bmatrix} = \frac{1}{8} \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & \frac{1+j}{\sqrt{2}} & j & -\frac{1+j}{\sqrt{2}} & -1 & \frac{1-j}{\sqrt{2}} & -j & \frac{1-j}{\sqrt{2}} \\ 1 & j & -1 & -j & 1 & j & -1 & -j \\ 1 & -\frac{1+j}{\sqrt{2}} & -j & \frac{1+j}{\sqrt{2}} & -1 & \frac{1-j}{\sqrt{2}} & j & -\frac{1-j}{\sqrt{2}} \\ 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 \\ 1 & -\frac{1-j}{\sqrt{2}} & j & \frac{1-j}{\sqrt{2}} & -1 & \frac{1+j}{\sqrt{2}} & -j & -\frac{1+j}{\sqrt{2}} \\ 1 & -j & -1 & j & 1 & -j & -1 & j \\ 1 & \frac{1-j}{\sqrt{2}} & -j & -\frac{1-j}{\sqrt{2}} & -1 & \frac{1+j}{\sqrt{2}} & j & \frac{1+j}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 28 \\ -4+j9.657 \\ -4+4j \\ -4+j1.657 \\ -4 \\ -4-j1.657 \\ -4-4j \\ -4-j9.657 \end{bmatrix}$$

$$\begin{aligned}
 &= \frac{1}{8} \left[\begin{aligned} &28 - 4 + j9.657 - 4 + 4j - 4 + j1.657 - 4 - 4 - j1.657 - 4 - 4j - 4 - j9.657 \\ &28 + \left(\frac{1+j}{\sqrt{2}}\right)(-4+j9.657) - 4j - 4 + \left(\frac{-1+j}{\sqrt{2}}\right)(-4+j1.657) + 4 + \\ &\quad \left(\frac{-1-j}{\sqrt{2}}\right)(-4-j1.657) + 4j - 4 + \left(\frac{1-j}{\sqrt{2}}\right)(-4-j9.657) \\ &28 - 4j - 9.657 + 4 - 4j + 4j + 1.657 - 4 - 4j + 1.657 + 4 + 4j + 4j \\ &\quad - 9.657 \\ &28 + \left(\frac{-1+j}{\sqrt{2}}\right)(-4+j9.657) + 4j + 4 + \left(\frac{1+j}{\sqrt{2}}\right)(-4+j1.657) + 4 \\ &\quad \left(\frac{1-j}{\sqrt{2}}\right)(-4-j1.657) - 4j + 4 + \left(\frac{-1-j}{\sqrt{2}}\right)(-4-j9.657) \\ &28 + 4 - j9.657 - 4 + 4j + 4 - j1.657 - 4 + 4 + j1.657 - 4 - 4j + 4 + j9.657 \\ &28 + \left(\frac{-1-j}{\sqrt{2}}\right)(-4+j9.657) - 4j - 4 + \left(\frac{1-j}{\sqrt{2}}\right)(-4+j1.657) + 4 + \left(\frac{1+j}{\sqrt{2}}\right)(-4-j1.657) \\ &\quad + 4j - 4 + \left(\frac{1+j}{\sqrt{2}}\right)(-4-j9.657) \\ &28 + 4j + 9.657 + 4 - 4j - 4j - 1.657 - 4 + 4j - 1.657 + 4 + 4j - 4j + 9.657 \\ &28 + \left(\frac{1-j}{\sqrt{2}}\right)(-4+j9.657) + 4j + 4 + \left(\frac{-1-j}{\sqrt{2}}\right)(-4+j1.657) + 4 + \\ &\quad \left(\frac{1+j}{\sqrt{2}}\right)(-4-j1.657) - 4j + 4 + \left(\frac{1+j}{\sqrt{2}}\right)(-4-j9.657) \end{aligned} \right]
 \end{aligned}$$

- 0
- 1
- 2
- 3
- 4
- 5
- 6
- 7

compute of NPT dft of the following discrete sequence $x(n) = \{2, 1, 2, 1, 1, 2, 1, 2\}$ using matrix method

Soln

$$\begin{bmatrix} x(0) \\ x(1) \\ x(2) \\ x(3) \\ x(4) \\ x(5) \\ x(6) \\ x(7) \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & \frac{1-j}{\sqrt{2}} & -j & \frac{1+j}{\sqrt{2}} & -1 & \frac{-1-j}{\sqrt{2}} & j & \frac{1+j}{\sqrt{2}} \\ 1 & -j & -1 & j & 1 & j & -1 & j \\ 1 & \frac{-1-j}{\sqrt{2}} & j & \frac{1-j}{\sqrt{2}} & -1 & \frac{1+j}{\sqrt{2}} & -j & \frac{-1+j}{\sqrt{2}} \\ 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 \\ 1 & \frac{1+j}{\sqrt{2}} & -j & \frac{1-j}{\sqrt{2}} & -1 & \frac{-1-j}{\sqrt{2}} & j & \frac{-1+j}{\sqrt{2}} \\ 1 & j & -1 & -j & 1 & j & -1 & -j \\ 1 & \frac{1-j}{\sqrt{2}} & j & \frac{-1+j}{\sqrt{2}} & -1 & \frac{1-j}{\sqrt{2}} & -j & \frac{1+j}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ 2 \\ 1 \\ 1 \\ 2 \\ 1 \\ 2 \end{bmatrix}$$

$$= \begin{bmatrix} 2+1+2+1+1+2+1+2 \\ 2+0.707-j0.707-2j-0.707-j0.707-1-j0.707+j0.707+j+2(0.707)j \\ 2-j-2+j+1-2j-1+2j \\ 2-0.707-j0.707+2j+0.707-j0.707-1+1.414+j1.414 \\ 2-1+j2-1+1-2+j1-2 \\ 2-1+j2-1+1-2+j1-2 \\ 2-j-1.414+j1.414 \\ 2-1+j2-1+1-2+j1-2 \end{bmatrix}$$

$$= \begin{bmatrix} 2 - 1.414 + j1.414 - 2j + 0.707 + j0.707 - 1 + 1.414 - j1.414 + \\ j - 1.414 - j1.414 \\ 2 + j - 2 - j + 1 + 2j - 1 - 2j \\ 2 + 0.707 + j0.707 + 2j - 0.707 + j0.707 - 1 - \\ 1.414 - j1.414 - j + 1.414 - j1.414 \end{bmatrix}$$

$$\begin{bmatrix} x(0) \\ x(1) \\ x(2) \\ x(3) \\ x(4) \\ x(5) \\ x(6) \\ x(7) \end{bmatrix} = \begin{bmatrix} 12 \\ 1 + 0.414j \\ 0 \\ 1 + 2.414j \\ 0 \\ 1 - 2.414j \\ 0 \\ 1 - 0.414j \end{bmatrix}$$

Determine N-pt DFT of the given sequence.

$$x(n) = \{1, 2, 1, 2, 1, 3, 1, 3\}$$

$$\begin{matrix} \text{sdn} \\ \begin{bmatrix} x(0) \\ x(1) \\ x(2) \\ x(3) \\ x(4) \\ x(5) \\ x(6) \\ x(7) \end{bmatrix} \end{matrix} = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & \frac{1-j}{\sqrt{2}} & -j & \frac{-1-j}{\sqrt{2}} & -1 & \frac{-1+j}{\sqrt{2}} & j & \frac{1+j}{\sqrt{2}} \\ 1 & -j & -1 & j & 1 & -j & -1 & j \\ 1 & \frac{-1-j}{\sqrt{2}} & j & \frac{1-j}{\sqrt{2}} & -1 & \frac{1+j}{\sqrt{2}} & -j & \frac{-1+j}{\sqrt{2}} \\ 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 \\ 1 & \frac{-1+j}{\sqrt{2}} & -j & \frac{1+j}{\sqrt{2}} & -1 & \frac{1-j}{\sqrt{2}} & j & \frac{-1-j}{\sqrt{2}} \\ 1 & j & -1 & -j & 1 & j & -1 & -j \\ 1 & \frac{1+j}{\sqrt{2}} & j & \frac{-1-j}{\sqrt{2}} & -1 & \frac{-1-j}{\sqrt{2}} & -j & \frac{1-j}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 1 \\ 2 \\ 1 \\ 3 \\ 1 \\ 3 \end{bmatrix}$$

$$\begin{aligned}
 & 1 + 2 + 1 + 2 + 1 + 3 + 1 + 3 \\
 = & \left[\begin{aligned}
 & 1 + 1.414 - j - 1.414 - j - 1.414 - j - 1.414 - 1 - 2.121 + j 2.121 \\
 & + j + 2.121 + j 2.121 \\
 & 1 - 2j - 1 + 2j + 1 - 3j - 1 + 3j \\
 & 1 - 1.414 - j - 1.414 + j + 1.414 - j - 1.414 - 1 + 2.121 + j 2.121 \\
 & - j - 2.121 + j 2.121 \\
 & 1 - 2 + 1 - 2 + 1 - 3 + 1 - 3 \\
 & 1 - 1.414 + j 1.414 - j + 1.414 + j 1.414 - 1 + 2.121 - j 2.121 \\
 & + j - 2.121 - j 2.121 \\
 & 1 + 2j - 1 - 2j + 1 + 3j - 1 - 3j \\
 & 1 + 1.414 + j 1.414 + j - 1.414 + j 1.414 - 1 - 2.121 - j 2.121 \\
 & - j + 2.121 - j 2.121
 \end{aligned} \right]
 \end{aligned}$$

$$\begin{bmatrix} x(0) \\ x(1) \\ x(2) \\ x(3) \\ x(4) \\ x(5) \\ x(6) \\ x(7) \end{bmatrix} = \begin{bmatrix} 14 \\ 1.414j \\ 0 \\ 1.414j \\ -6 \\ -1.414j \\ +j \\ -1.414j \end{bmatrix}$$

circular convolution (diff bet circular & linear)

17/26. Properties
 periodic signal (after some period, the same sequence is to be repeated)

$$\{1, -1, 1, -1\} \quad \{1, -1, 1, -1\} \rightarrow \text{periodic sequence}$$

$N=4$
 $N=4$
 fundamental period of discrete sequence.

10/11

circular
 information cannot be obtained from the signal

→ sinusoidal shift can be the sequence periodic

$$x(n+N) = x(n)$$

$$-3 \bmod 4 = 1$$

$$x(-n \bmod N) = x(N-n)$$

5. complex conjugate property of 1D DFT

$$x(n) \xrightarrow[\text{DFT}]{N\text{pt}} X(k)$$

$$x^*(n) \longrightarrow X^*(N-k)$$

$$X(k) = \sum_{n=0}^{N-1} x(n) e^{-j \frac{2\pi}{N} nk}$$

$$X(k) = \sum_{n=0}^{N-1} [x^*(n)] e^{-j \frac{2\pi}{N} nk}$$

$$= \left[\sum_{n=0}^{N-1} x(n) e^{j \frac{2\pi}{N} nk} \right]^*$$

$$= \sum_{n=0}^{N-1} \left[x(n) e^{j \frac{2\pi}{N} nk} \right]^* \quad (1)$$

$$= \sum_{n=0}^{N-1} \left[x(n) e^{j \frac{2\pi}{N} nk} \cdot e^{-j \frac{2\pi}{N} nN} \right]^*$$

$$= \sum_{n=0}^{N-1} \left[x(n) e^{-j \frac{2\pi}{N} (N-k)n} \right]^*$$

$$= X^*(N-k)$$

6. circularly frequency shift property

$$x(n) \xrightarrow{\quad} x(k)$$

$$x(n) e^{j \frac{2\pi}{N} n m} \xrightarrow{\quad} x((k-m))_N$$

$$x(k) = \sum_{n=0}^{N-1} x(n) e^{-j \frac{2\pi}{N} n k}$$

$$= \sum_{n=0}^{N-1} x(n) e^{j \frac{2\pi}{N} n m} e^{-j \frac{2\pi}{N} n k}$$

$$= \sum_{n=0}^{N-1} x(n) e^{-j \frac{2\pi}{N} (k-m) n}$$

$$= x((k-m))_N \quad \text{N.} \rightarrow \text{DFT should be taken N times}$$

7. multiplication property

$$x_1(n) \xrightarrow[\text{DFT}]{N\text{-pt}} X_1(k)$$

$$x_2(n) \xrightarrow[\text{DFT}]{N\text{-pt}} X_2(k)$$

$$x_1(n) x_2(n) \xrightarrow[\text{DFT}]{N\text{-pt}} \frac{1}{N} \left[X_1(k) \otimes X_2(k) \right]$$

(circular convolution)

$$= \sum_{n=0}^{N-1} [x_1(n) x_2(n)] e^{-j \frac{2\pi}{N} n k}$$

instead of $x_1(n)$ sub its IDFT of $x_1(k)$.

$$\text{u/KT, } x(n) = \sum_{k=0}^{N-1} X(k) e^{j \frac{2\pi}{N} n k}$$

$$= \sum_{n=0}^{N-1} \left[\frac{1}{N} \sum_{m=0}^{N-1} X_1(m) e^{j \frac{2\pi}{N} n m} \right] x_2(n) e^{-j \frac{2\pi}{N} n k}$$

$$\begin{aligned}
&= \frac{1}{N} \sum_{m=0}^{N-1} x_1(m) \sum_{n=0}^{N-1} x_2(n) e^{j\frac{2\pi}{N}nm} e^{-j\frac{2\pi}{N}nk} \\
&= \frac{1}{N} \sum_{m=0}^{N-1} x_1(m) \left[\sum_{n=0}^{N-1} x_2(n) e^{j\frac{2\pi}{N}nm} \cdot e^{-j\frac{2\pi}{N}nk} \right] \\
&= \frac{1}{N} \sum_{m=0}^{N-1} x_1(m) \left[\sum_{n=0}^{N-1} x_2(n) e^{-j\frac{2\pi}{N}(k-m)n} \right] \\
&= \frac{1}{N} \sum_{m=0}^{N-1} x_1(m) x_2((k-m))_N
\end{aligned}$$

Next.

Linear convolution

$$x_1(n) * x_2(n) = \sum_{m=-\infty}^{+\infty} x_1(m) x_2(n-m)$$

Circular convolution

$$x_1(n) \otimes x_2(n) = \sum_{m=0}^{N-1} x_1(m) x_2((n-m))_N$$

Therefore

$$= \frac{1}{N} x_1(k) \otimes x_2(k) \quad \text{by problem}$$

Linear convolution
both sequences
may equal or
unequal

Circular convolu

but both sequences
must be equal otherwise we have to
make it equal

Eg. $x_1 \rightarrow 5$ } which can be used
 $x_2 \rightarrow 3$ } $5-3$ or $3+2$

$$N = \max(N_1, N_2)$$

Therefore make $x_2(n)$ will
adjust to length of x_1

Ex: max len is $x_1 \rightarrow 5$

Prop of N-pt DFT [continuation]

① Linearity

$$x_1(n) \xrightarrow[\text{DFT}]{\text{N-pt}} x_1(k)$$

$$a_1 x_1(n) \xrightarrow[\text{DFT}]{\text{N-pt}} a_1 x_1(k)$$

$$a_2 x_2(n) \longrightarrow a_2 x_2(k)$$

$$[a_1 x_1(n) + a_2 x_2(n)] \xrightarrow[\text{DFT}]{\text{N-pt}} a_1 [x_1(k)] + a_2 [x_2(k)]$$

Proof

$$\text{N-pt DFT}[x_1(n)] = \sum_{n=0}^{N-1} x_1(n) e^{-j\frac{2\pi}{N}nk} \longrightarrow \textcircled{1}$$

$$\text{N-pt DFT}[a_1 x_1(n)] = a_1 \cdot \sum_{n=0}^{N-1} x_1(n) e^{-j\frac{2\pi}{N}nk}$$

$$\text{N-pt DFT}[a_1 x_1(n)] = a_1 x_1(k) \longrightarrow \textcircled{2}$$

$$\text{Similarly, N-pt DFT}[a_2 x_2(n)] = a_2 x_2(k) \longrightarrow \textcircled{3}$$

$$\text{N-pt DFT}[a_1 x_1(n) + a_2 x_2(n)] = \sum_{n=0}^{N-1} [a_1 x_1(n) + a_2 x_2(n)] e^{-j\frac{2\pi}{N}nk}$$

$$a_1 x_1(n) + a_2 x_2(n) \xrightarrow[\text{DFT}]{\text{N-pt}} \sum_{n=0}^{N-1} a_1 x_1(n) e^{-j\frac{2\pi}{N}nk} + \sum_{n=0}^{N-1} a_2 x_2(n) e^{-j\frac{2\pi}{N}nk}$$

$$a_1 x_1(n) + a_2 x_2(n) \xrightarrow[\text{DFT}]{\text{N-pt}} a_1 \sum_{n=0}^{N-1} x_1(n) e^{-j\frac{2\pi}{N}nk} + a_2 \sum_{n=0}^{N-1} x_2(n) e^{-j\frac{2\pi}{N}nk}$$

$$a_1 x_1(n) + a_2 x_2(n) \xrightarrow[\text{DFT}]{\text{N-pt}} a_1 x_1(k) + a_2 x_2(k)$$

∴ Hence proved.

② Periodicity:

$$x(n) \xrightarrow[\text{DFT}]{\text{N-pt}} x(k)$$

$$x(n+N) \xrightarrow[\text{DFT}]{\text{N-pt}} x(k+N)$$

$$x(n) = x(n+N) \quad \forall n$$

N → Fundamental period

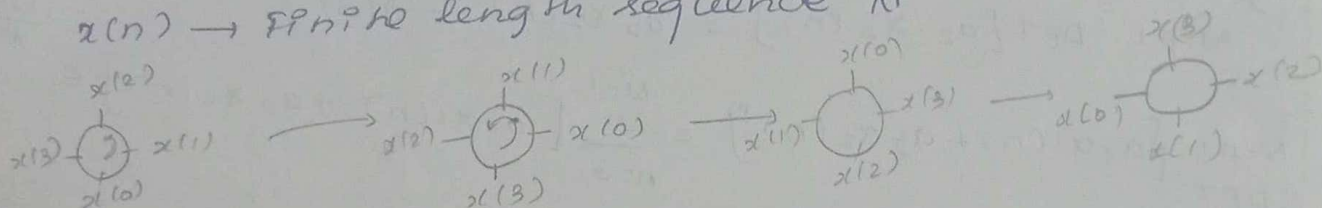
$$X(K) = N\text{-pt DFT of } 'x(n)' = \sum_{n=0}^{N-1} x(n) e^{-\frac{j2\pi}{N} nk}$$

$$\begin{aligned} X(K+N) &= \sum_{n=0}^{N-1} x(n) e^{-\frac{j2\pi}{N} n(K+N)} \\ &= \sum_{n=0}^{N-1} x(n) e^{-\frac{j2\pi}{N} nk} \cdot e^{-\frac{j2\pi}{N} nN} \\ &= \sum_{n=0}^{N-1} x(n) e^{-\frac{j2\pi}{N} nk} \cdot e^{-j2\pi n} \cdot e^{+jn\pi} = 1 \\ &= \sum_{n=0}^{N-1} x(n) e^{-\frac{j2\pi}{N} nk} \quad (1) \end{aligned}$$

$$X(K+N) = X(K)$$

③ Circular Time Shift

$x(n) \rightarrow$ Finite length sequence 'N'



$$x(n) \xrightarrow[N\text{-pt DFT}]{N\text{-pt}} X(K)$$

$$x((n-m))_N \xrightarrow[N\text{-pt DFT}]{N\text{-pt}} X(K) e^{-\frac{j2\pi}{N} Km}$$

$x((n-m))_N \xrightarrow{\text{Time shifted version of } x(n)}$
 $x((n-m) \bmod N)$
 (Note: $\text{mod } N$ is indicated by a curved arrow pointing to the expression)

Proof

$$\text{DFT}[x(n)] = \sum_{n=0}^{N-1} x(n) e^{-\frac{j2\pi}{N} nk}$$

$$\therefore \text{DFT}[x((n-m))_N] = \sum_{n=0}^{N-1} x((n-m))_N e^{-\frac{j2\pi}{N} nk}$$

let, $p = n - m \Rightarrow n = p + m$

if $n = 0 \Rightarrow p = m$

$n = N - 1 \Rightarrow p = N - 1 - m$

$$\begin{aligned}
 [x((n-m)_N)] &\xrightarrow[\text{DFT}]{N-pt} \sum_{p=m}^{N-1-m} x(p) e^{-\frac{j2\pi}{N} (p+m)k} \\
 &= \sum_{p=0}^{N-1} x(p) e^{-\frac{j2\pi}{N} pk} e^{-\frac{j2\pi}{N} mk} \\
 &= e^{-\frac{j2\pi}{N} mk} \sum_{p=0}^{N-1} x(p) e^{-\frac{j2\pi}{N} pk}
 \end{aligned}$$

$$x((n-m)_N) \xrightarrow[\text{DFT}]{N-pt} e^{-\frac{j2\pi}{N} mk} \cdot X(k)$$

Time Reversal

$$x(n) \rightarrow x(-n)$$

$$x(n) \xrightarrow[\text{DFT}]{N-pt} X(k)$$

$$x(N-n) \xrightarrow[\text{DFT}]{N-pt} X(N-k)$$

Proof: $N\text{-pt DFT}[x(n)] = X(k) = \sum_{n=0}^{N-1} x(n) e^{-\frac{j2\pi}{N} nk} \rightarrow \textcircled{1}$

$$\text{DFT}[x(N-n)] = \sum_{n=0}^{N-1} x(N-n) e^{-\frac{j2\pi}{N} nk}$$

$$\text{Let, } m = N-n \Rightarrow n = N-m$$

$$\begin{cases} \text{if } n=0, m=N \\ \text{if } n=N-1, m=1 \end{cases}$$

$$= \sum_{m=N}^1 x(m) e^{-\frac{j2\pi}{N} (N-m)k}$$

$$= \sum_{m=0}^{N-1} x(m) e^{\frac{j2\pi}{N} mk} \cdot e^{-\frac{j2\pi}{N} Nk}$$

$$= \sum_{m=0}^{N-1} x(m) e^{\frac{j2\pi}{N} mk} \cdot (1)$$

$$= \sum_{m=0}^{N-1} x(m) e^{\frac{j2\pi}{N} mk} \cdot e^{-\frac{j2\pi}{N} Nm}$$

$$= \sum_{m=0}^{N-1} x(m) e^{-\frac{j2\pi}{N} (N-k)m}$$

$$x(N-n) \xrightarrow[\text{DFT}]{N-Pt} x(N-k)$$

∴ Hence proved.

Circular convolution

$$x_1(n) \longrightarrow N_1 \text{ (length)}$$

$$x_2(n) \longrightarrow N_2$$

$$x_1(n) \oplus x_2(n) = N = \max(N_1, N_2)$$

length of two sequences must be equal.

$$N_1 = N_2$$

mathematical exp

$$y(n) = \sum_{k=0}^{N-1} x_1(m) x_2(k-m) \quad N.$$

sup = 0

Linear convolution

$$x_1(n) \longrightarrow N_1$$

$$x_2(n) \longrightarrow N_2$$

$$x_1(n) * x_2(n) \longrightarrow N = N_1 + N_2 - 1$$

length of two sequences may or may not be have equal length

$$N_1 \neq N_2 \quad N_1 = N_2$$

mathematical exp

$$y(n) = \sum_{k=-\infty}^{\infty} x_1(k) x_2(n-k)$$

Tabular method (matrix method)

$$N = 4$$

o/p matrix

$$\begin{bmatrix} y(0) \\ y(1) \\ y(2) \\ y(3) \end{bmatrix} = \begin{bmatrix} x_2(0) & x_2(3) & x_2(2) & x_2(1) \\ x_2(1) & x_2(0) & x_2(3) & x_2(2) \\ x_2(2) & x_2(1) & x_2(0) & x_2(3) \\ x_2(3) & x_2(2) & x_2(1) & x_2(0) \end{bmatrix} \begin{bmatrix} x_1(0) \\ x_1(1) \\ x_1(2) \\ x_1(3) \end{bmatrix}$$

max(N₁, N₂) = N × 1
4 × 1

shift
N × N
4 × 4

linear
N × 1
4 × 1

$$x(0), x(1), x(2), x(3)$$

$$x(3), x(0), x(1), x(2)$$

$$x(2), x(3), x(0), x(1)$$

$$x(1), x(2), x(3), x(0)$$

Perform circular convolution between the following two discrete sequences,

$$x_1(n) = \{1, 2, 3, 4\}$$

$$x_2(n) = \{5, 6, 7, 8\}$$

using matrix or Tabular method.

Soln

$$N_1 = N_2 = 4.$$

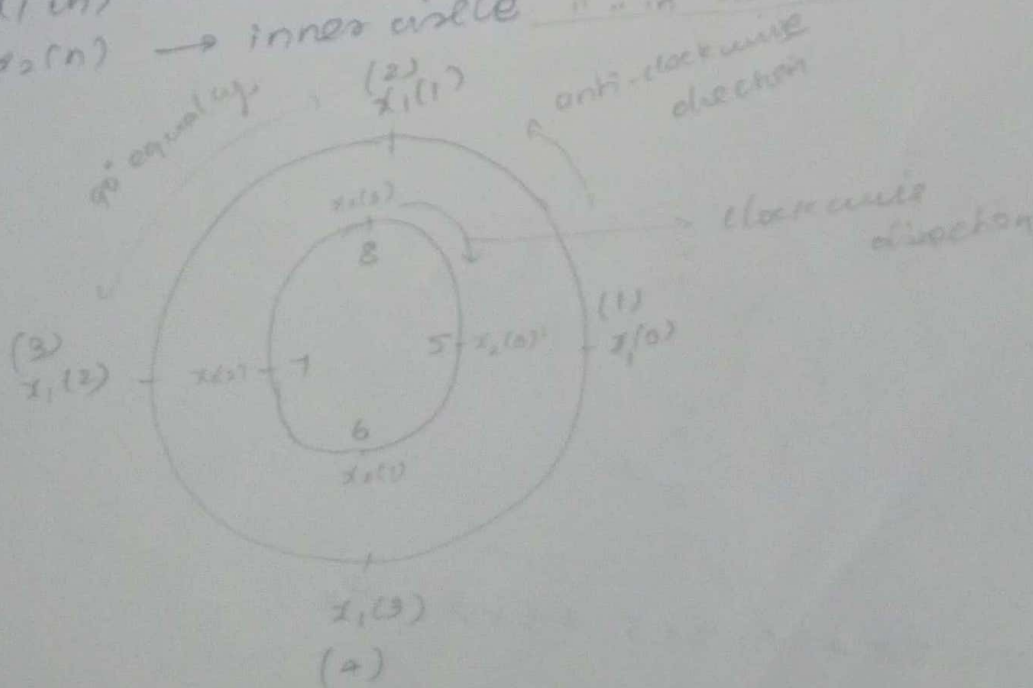
$$\begin{bmatrix} y(0) \\ y(1) \\ y(2) \\ y(3) \end{bmatrix}_{4 \times 1} = \begin{bmatrix} 5 & 8 & 7 & 6 \\ 6 & 5 & 8 & 7 \\ 7 & 6 & 5 & 8 \\ 8 & 7 & 6 & 5 \end{bmatrix}_{4 \times 4} \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}_{4 \times 1}$$

$$\begin{bmatrix} y(0) \\ y(1) \\ y(2) \\ y(3) \end{bmatrix} = \begin{bmatrix} 5 + 16 + 21 + 24 \\ 6 + 10 + 24 + 28 \\ 7 + 12 + 15 + 22 \\ 8 + 14 + 18 + 20 \end{bmatrix} = \begin{bmatrix} 66 \\ 68 \\ 66 \\ 60 \end{bmatrix}$$

$$y(n) = \{66, 68, 66, 60\}, N = 4.$$

By Graphical method

$x_1(n) \rightarrow$ outer circle
 $x_2(n) \rightarrow$ inner circle



$$y(n) = \sum_{m=0}^{N-1} x_1(m) x_2(n-m) \quad \text{Time reversed}$$

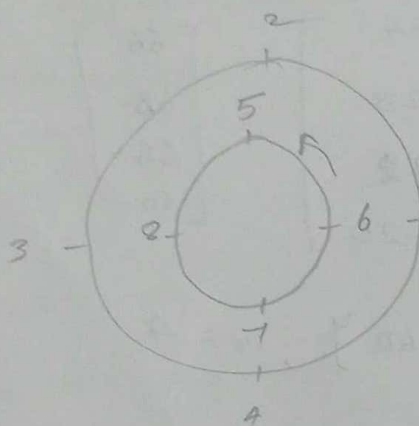
Step 1: $n=0$, $y(0) = \sum_{m=0}^{N-1} x_1(m) x_2(10-m)$

$$y(0) = (5 \times 1) + (6 \times 4) + (7 \times 3) + (8 \times 2)$$

$$= 5 + 24 + 21 + 16$$

$$= 66$$

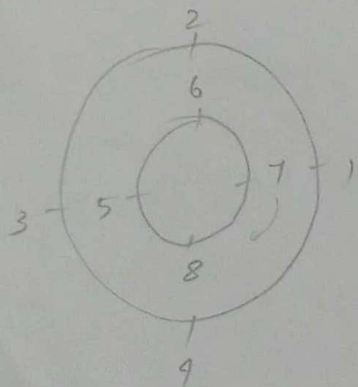
Step 2: keep outer circle fixed, inner circle shift anticlockwise



$$y(1) = 6 \times 1 + 28 + 24 + 10$$

$$y(1) = 68$$

Step 3:

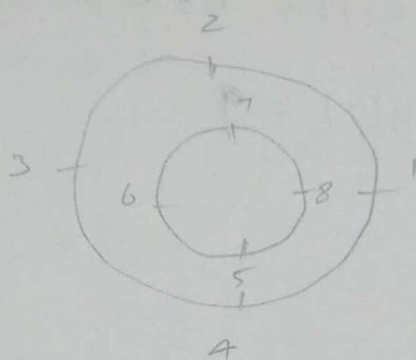


$$y(2) = 8 \times 4 + 3 \times 5 + 6 \times 2 + 7 \times 1$$

$$= 32 + 15 + 12 + 7$$

$$y(2) = 66$$

Step 4:



$$y(3) = 8 \times 1 + 5 \times 4 + 3 \times 6 + 7 \times 2$$

$$= 8 + 20 + 18 + 14$$

$$y(3) = 60$$

$$y(n) = \{66, 68, 66, 60\}$$

Perform circular convolution for following discrete sequences using graphical method.

$$x_1(n) = \{1, -1, 2\}$$

$$x_2(n) = \{-1, 0, 1, 4\}$$

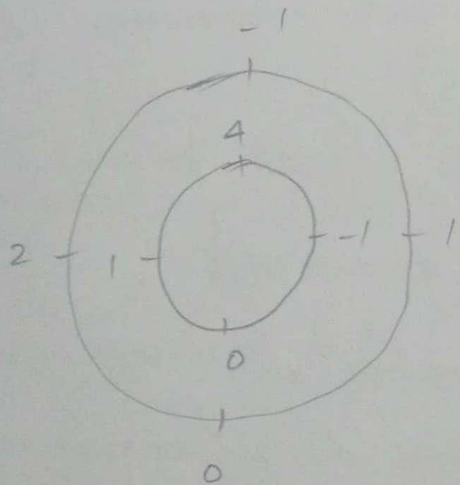
$$N_1 \neq N_2$$

$$x_1(n) = \{1, -1, 2, 0\}$$

$$N_1 = 4$$

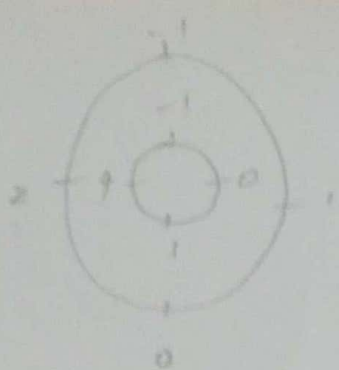
$$x_2(n) = \{-1, 0, 1, 4\}$$

$$N_2 = 4$$



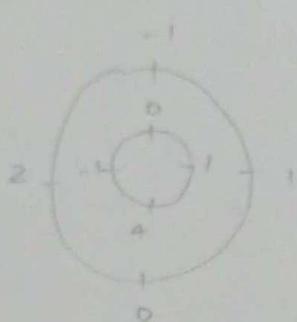
$$y(0) = -1 + 0 + 2 + 4$$

$$y(0) = -5$$



$$y(1) = 0 + 0 + 8 + 1$$

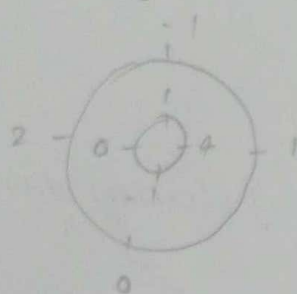
$$y(1) = 9$$



$$y(2) = 1 + 0 + -2 + 0$$

$$= 1 - 2$$

$$= -1$$



$$y(3) = 4 + 0 + 0 - 1$$

$$y(3) = 3$$

$$y(n) = \{-3, 9, -1, 3\}$$

$$\begin{bmatrix} -1 & 4 & 1 & 0 \\ 0 & -1 & 4 & 1 \\ 1 & 0 & -1 & 4 \\ 4 & 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 2 \\ 0 \end{bmatrix} = \begin{bmatrix} -3 \\ 9 \\ -1 \\ 3 \end{bmatrix}$$

how to determine LC by means of CC

$$N_1 = 5$$

$$N_2 = 3$$

$$LC \rightarrow N = 5 + 3 - 1 = 7 \quad CC \rightarrow N = 5$$

total ^{each} sequence must be in equal, in such a way that

make N_1 & N_2 (7) by append zero

$$\begin{array}{l} N_1 = 5 \xrightarrow{00} 7 \rightarrow \underline{(N_2 - 1)} \text{ zeros at 1st sequence} \\ N_2 = 3 \xrightarrow{0000} 7 \rightarrow \underline{(N_1 - 1)} \text{ zeros at 2nd sequence} \end{array}$$

compute the linear convolution bet the following two discrete sequences

$$x(n) = \{1, -1\}$$

$$h(n) = \{1, 2, -1\}$$

using circular convolution

[i.e. → to find the response of linear system]

soln

Given;

$$N_1 \rightarrow 2$$

$$N_2 \rightarrow 3$$

$(N_2 - 1)$ zeros at 1st sequence
 $3 - 1 = 2$

$$x(n) = \{1, -1, 0, 0\}$$

$(N_1 - 1)$ zeros at 2nd sequence
 $2 - 1 = 1$

$$h(n) = \{1, 2, -1, 0\}$$

By Linear convolution using tabular method

$h(n) \backslash x(n)$	1	-1
1	1	-1
2	2	-2
-1	-1	1

$$y(n) = \{1, 1, -3, 1\}$$

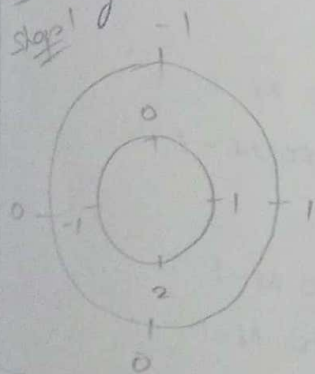
$$N = 4$$

$$N = N_1 + N_2 - 1$$

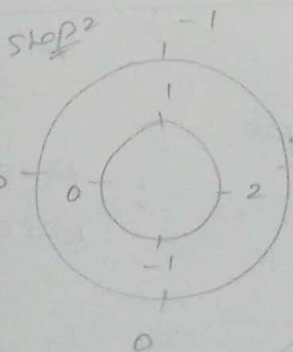
$$N = 2 + 3 - 1$$

$$N = 4$$

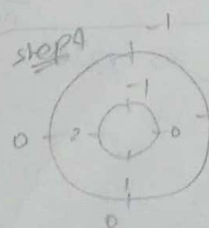
→ By ~~convolve~~ circular convolution method



$$y(0) = 1 + 0 + 0 + 0 = 1$$



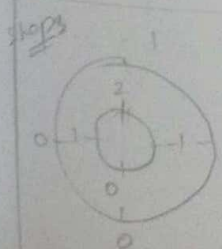
$$y(1) = 2 + 0 + 0 - 1 = 1$$



$$y(2) = 0 + 0 + 0 + 1 = 1$$

$$y(n) = \{1, 1, -3, 1\}$$

$$N = 4$$



$$y(2) = -1 + 0 + 0 - 2 = -3$$

By Tabular method [verification]

$$\begin{bmatrix} y(0) \\ y(1) \\ y(2) \\ y(3) \end{bmatrix} = \begin{bmatrix} 1 & 0 & -1 & 2 \\ 2 & 1 & 0 & -1 \\ -1 & 2 & 1 & 0 \\ 0 & -1 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 0 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1+0+0+0 \\ 2-1+0+0 \\ -1-2+0+0 \\ 0+1+0+0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ -3 \\ +1 \end{bmatrix}$$

$$y(n) = \{1, 1, -3, 1\}$$

circular convolution pply

(16m) By using the pply of
N-pt DFT the in
term of CC

$$x_1(n) \otimes x_2(n) \xrightarrow[\text{DFT}]{\text{N-pt}} x_1(k) \cdot x_2(k)$$

$$x_1(n) \xrightarrow[\text{DFT}]{\text{N-pt}} x_1(k)$$

$$x_2(n) \xrightarrow[\text{DFT}]{\text{N-pt}} x_2(k)$$

$$\text{here, } x_1(n) \otimes x_2(n) = \sum_{m=0}^{N-1} x_1(m) x_2((n-m))_N$$

circular convolution
bet 2 sequences

$$\text{Let } x_1(k) = \sum_{m=0}^{N-1} x_1(m) e^{-j \frac{2\pi}{N} mk} \quad ; \quad \begin{matrix} k \rightarrow 0 \text{ to } N-1 \\ m \rightarrow 0 \text{ to } N-1 \end{matrix}$$

$$x_2(k) = \sum_{p=0}^{N-1} x_2(p) e^{-j \frac{2\pi}{N} pk} \quad ; \quad \begin{matrix} p \rightarrow 0 \text{ to } N-1 \\ k \rightarrow 0 \text{ to } N-1 \end{matrix}$$

of

$$\text{N-pt DFT} [x_1(n) \otimes x_2(n)] = x_1(k) \cdot x_2(k)$$

then,

$$N\text{-pt IDFT}[x_1(k) x_2(k)] = x_1(n) * x_2(n)$$

Proof WKT,

$$N\text{-pt IDFT}[x(k)] = \frac{1}{N} \sum_{k=0}^{N-1} x(k) e^{\frac{j2\pi}{N} nk}$$

$$N\text{-pt IDFT}[x_1(k) \cdot x_2(k)] = \frac{1}{N} \sum_{k=0}^{N-1} x_1(k) x_2(k) e^{\frac{j2\pi}{N} nk}$$

$$= \frac{1}{N} \sum_{k=0}^{N-1} \left[\sum_{m=0}^{N-1} x_1(m) e^{-\frac{j2\pi}{N} mk} \right] \left[\sum_{p=0}^{N-1} x_2(p) e^{-\frac{j2\pi}{N} pk} \right] e^{\frac{j2\pi}{N} nk}$$

$$= \frac{1}{N} \sum_{m=0}^{N-1} x_1(m) \sum_{k=0}^{N-1} \sum_{p=0}^{N-1} x_2(p) e^{-\frac{j2\pi}{N} mk} \cdot e^{-\frac{j2\pi}{N} pk} e^{\frac{j2\pi}{N} nk}$$

$$= \frac{1}{N} \sum_{m=0}^{N-1} x_1(m) \sum_{k=0}^{N-1} \sum_{p=0}^{N-1} x_2(p) e^{\frac{j2\pi}{N} k(n-p-m)}$$

$$= \frac{1}{N} \sum_{m=0}^{N-1} x_1(m) \sum_{p=0}^{N-1} x_2(p) \sum_{k=0}^{N-1} e^{\frac{j2\pi}{N} k(n-p-m)} \quad \rightarrow \textcircled{1}$$

let $n-p-m = qN \Rightarrow p = \boxed{n-m-qN}$

$$\sum_{k=0}^{N-1} e^{\frac{j2\pi}{N} k(n-p-m)} = \sum_{k=0}^{N-1} e^{\frac{j2\pi}{N} qN k}$$

$$= \sum_{k=0}^{N-1} \left(e^{j2\pi q} \right)^k$$

$$= \sum_{k=0}^{N-1} (1)^k$$

$$\sum_{k=0}^{N-1} e^{\frac{j2\pi}{N} k(n-p-m)} = N$$

sub in eqn $\textcircled{1}$

$\textcircled{1} \Rightarrow$

$$= \frac{1}{N} \sum_{m=0}^{N-1} x_1(m) \sum_{p=0}^{N-1} x_2(p) \sum_{k=0}^{N-1} e^{\frac{j2\pi}{N} k(n-p-m)}$$

$$= \frac{1}{N} \sum_{m=0}^{N-1} x_1(m) \sum_{p=0}^{N-1} x_2(p) (N)$$

$$N\text{-pt IDFT}[x_1(k)x_2(k)] = \sum_{m=0}^{N-1} x_1(m) \sum_{p=0}^{N-1} x_2(p) \rightarrow (2)$$

$$\begin{aligned} \sum_{p=0}^{N-1} x_2(p) &= \sum_{p=0}^{N-1} x_2(n-m-p) \\ &= \sum_{p=0}^{N-1} x_2((n-m))_N \end{aligned}$$

$$= \sum_{m=0}^{N-1} x_2((n-m))_N$$

$$\begin{aligned} \therefore N\text{-pt IDFT}[x_1(k)x_2(k)] &= \sum_{m=0}^{N-1} x_1(m) \sum_{n=0}^{N-1} x_2((n-m))_N \\ &= \sum_{m=0}^{N-1} x_1(m) x_2((1-m))_N \end{aligned}$$

$$N\text{-pt IDFT}[x_1(k)x_2(k)] = x_1(n) \otimes x_2(n)$$

$$x_1(k)x_2(k) \xleftarrow{N\text{-pt DFT}}$$

Parseval's theorem

$$\text{If } N\text{-pt DFT}[x_1(n)] = X_1(k)$$

$$N\text{-pt DFT}[x_2(n)] = X_2(k), \text{ then}$$

$$\sum_{n=0}^{N-1} x_1(n) x_2^*(n) = \frac{1}{N} \sum_{k=0}^{N-1} X_1(k) X_2^*(k)$$

Proof

$$\begin{aligned} \frac{1}{N} \sum_{k=0}^{N-1} \underbrace{x_1(k) x_2^*(k)}_{\substack{\downarrow \\ N\text{-pt} \\ \text{DFT}}} &= \frac{1}{N} \sum_{k=0}^{N-1} \left[\sum_{n=0}^{N-1} x_1(n) e^{-j \frac{2\pi}{N} nk} \right] x_2^*(k) \\ &= \frac{1}{N} \sum_{n=0}^{N-1} x_1(n) \sum_{k=0}^{N-1} x_2^*(k) e^{-j \frac{2\pi}{N} nk} \end{aligned}$$

$$= \sum_{n=0}^{N-1} x_1(n) \left[\frac{1}{N} \sum_{k=0}^{N-1} x_2^*(k) e^{j\frac{2\pi}{N}nk} \right]^*$$

$$= \sum_{n=0}^{N-1} x_1(n) x_2^*(n)$$

$$\frac{1}{N} \sum_{k=0}^{N-1} x_1(k) x_2^*(k) = \sum_{n=0}^{N-1} x_1(n) x_2^*(n)$$

$$\text{if } x_1(n) = x_2(n)$$

$$\frac{1}{N} \sum_{k=0}^{N-1} x_1(k) x_1^*(k) = \sum_{n=0}^{N-1} x_1(n) x_1^*(n)$$

$$\boxed{\frac{1}{N} \sum_{k=0}^{N-1} |x(k)|^2 = \sum_{n=0}^{N-1} x_1^2(n)}$$

6/7/26

1. using the properties of N-point DFT and IDFT, compute the o/p response of the LTI system for the following i/p and impulse response,

$$x(n) = \{1, 2, 3\}, \quad h(n) = \{-1, 2\}$$

soln using circular convolution property:

If $x_1(n) \xrightarrow{\text{DFT}} x_1(k)$ and $x_2(n) \xrightarrow{\text{DFT}} x_2(k)$, then the N-pt circular convolution of $x_1(n)$ and $x_2(n)$ is represented in the frequency domain by the product of their DFTs:

$$x_1(n) \otimes x_2(n) \xrightarrow{\text{DFT}} x_1(k) x_2(k)$$

Equivalently:

$$\text{IDFT} \{x_1(k) x_2(k)\} = x_1(n) \otimes x_2(n)$$

length of i/p sequence, $N_1 = 3$

length of impulse response, $N_2 = 2$

The length of linear convolution is

$$N = N_1 + N_2 - 1$$

$$N = 3 + 2 - 1 = 4$$

compute 4pt DFT of $x(n)$

$$x(n) = \{1, 2, 3, 0\} \quad \& \quad h(n) = \{-1, 2, 0, 0\}$$

$$X = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \\ 0 \end{bmatrix} = \begin{bmatrix} 1+2+3+0 \\ 1-2j-3+0 \\ 1-2+3+0 \\ 1+2j-3+0 \end{bmatrix} = \begin{bmatrix} 6 \\ -2-2j \\ 2 \\ -2+2j \end{bmatrix}$$

compute DFT of $h(n)$

$$H = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \begin{bmatrix} -1 \\ 2 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} -1+2 \\ -1-2j \\ -1-2 \\ -1+2j \end{bmatrix} = \begin{bmatrix} 1 \\ -1-2j \\ -3 \\ -1+2j \end{bmatrix}$$

multiply the DFTs

$$Y(k) = X(k)H(k) = \begin{bmatrix} 6 \\ -2-2j \\ 2 \\ -2+2j \end{bmatrix} \begin{bmatrix} 1 \\ -1-2j \\ -3 \\ -1+2j \end{bmatrix} = \begin{bmatrix} 6 \\ -2+6j \\ -6 \\ -2-6j \end{bmatrix}$$

compute the IDFT

$$W = \frac{1}{4} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & j & -1 & -j \\ 1 & -1 & 1 & -1 \\ 1 & -j & -1 & j \end{bmatrix} \begin{bmatrix} 6 \\ -2+6j \\ -6 \\ -2-6j \end{bmatrix}$$

$$W = \frac{1}{4} \begin{bmatrix} 6-2+6j-6-2-6j \\ 6-2j-6+6+2j-6 \\ 6+2-6j-6+2+6j \\ 6+2j+6+6-2j+6 \end{bmatrix} = \frac{1}{4} \begin{bmatrix} -4 \\ 0 \\ 4 \\ 24 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \\ 1 \\ 6 \end{bmatrix}$$

$$y(n) = \{-1, 0, 1, 6\}$$

3. using the properties of N-pt (DFT) and IDFT, compute the response of the LTI system for the following i/p and impulse response

$$x(n) = \{1, 2, 1, 0\}, h(n) = \{2, -1, 3\}$$

soln

$$N_1 = 4, N_2 = 3; N = N_1 + N_2 - 1$$

$$= 4 + 3 - 1$$

$$N = 6$$

$$\therefore x(n) = \{1, 2, 1, 0, 0, 0\} \text{ \& } h(n) = \{2, -1, 3, 0, 0, 0\}$$

compute DFT of $x(n)$

$$x_1(n) \otimes x_2(n) \xrightarrow{\text{DFT}} X_1(K) X_2(K)$$

6-pt DFT matrix

$$\begin{bmatrix} W_6^0 & W_6^0 & W_6^0 & W_6^0 & W_6^0 & W_6^0 \\ W_6^0 & W_6^1 & W_6^2 & W_6^3 & W_6^4 & W_6^5 \\ W_6^0 & W_6^2 & W_6^4 & W_6^6 & W_6^8 & W_6^{10} \\ W_6^0 & W_6^3 & W_6^6 & W_6^9 & W_6^{12} & W_6^{15} \\ W_6^0 & W_6^4 & W_6^8 & W_6^{12} & W_6^{16} & W_6^{20} \\ W_6^0 & W_6^5 & W_6^{10} & W_6^{15} & W_6^{20} & W_6^{25} \end{bmatrix}$$

$$W_N^K = e^{-\frac{j2\pi}{N}nk}; W_6^0 = 1, W_6^6 = 1$$

$$W_6^1 = e^{-\frac{j2\pi}{6} \cdot 1} = e^{-j\pi/3} = \cos \pi/3 - j \sin \pi/3 = \frac{1}{2} - j \frac{\sqrt{3}}{2}$$

$$W_6^2 = e^{-\frac{j2\pi}{6} \cdot 2} = e^{-j2\pi/3} = \cos \frac{2\pi}{3} - j \sin \frac{2\pi}{3} = -\frac{1}{2} - j \frac{\sqrt{3}}{2}$$

$$W_6^3 = e^{-\frac{j2\pi}{6} \cdot 3} = e^{-j\pi} = \cos \pi - j \sin \pi = -1$$

$$W_6^4 = e^{-\frac{j2\pi}{6} \cdot 4} = e^{-j4\pi/3} = \cos \frac{4\pi}{3} - j \sin \frac{4\pi}{3} = -\frac{1}{2} + j \frac{\sqrt{3}}{2}$$

$$W_6^5 = e^{-\frac{j2\pi}{6} \cdot 5} = e^{-j5\pi/3} = \cos \frac{5\pi}{3} - j \sin \frac{5\pi}{3} = \frac{1}{2} + j \frac{\sqrt{3}}{2}$$

$$W_6^8 = W_6^6 \cdot W_6^2 = 1 \left(-\frac{1}{2} - j\frac{\sqrt{3}}{2} \right) = -\frac{1}{2} - j\frac{\sqrt{3}}{2}$$

$$W_6^{10} = W_6^6 \cdot W_6^4 = 1 \left(-\frac{1}{2} + j\frac{\sqrt{3}}{2} \right) = -\frac{1}{2} + j\frac{\sqrt{3}}{2}$$

$$W_6^9 = W_6^6 \cdot W_6^3 = 1(-1) = -1$$

$$W_6^{12} = W_6^6 \cdot W_6^6 = 1$$

$$W_6^{15} = W_6^6 \cdot W_6^9 = 1(-1) = -1$$

$$W_6^{16} = W_6^8 \cdot W_6^{10} = 1 \left(-\frac{1}{2} + j\frac{\sqrt{3}}{2} \right) = -\frac{1}{2} + j\frac{\sqrt{3}}{2}$$

$$W_6^{20} = W_6^8 \cdot W_6^{12} = -\frac{1}{2} - j\frac{\sqrt{3}}{2}$$

$$W_6^{25} = W_6^{10} \cdot W_6^{15} = \left(-\frac{1}{2} + j\frac{\sqrt{3}}{2} \right) (-1) = \frac{1}{2} - j\frac{\sqrt{3}}{2}$$

$$X_1(K) = \begin{bmatrix} 1 & 1 & 1 & 1 & \cancel{\frac{1}{2} + j\frac{\sqrt{3}}{2}} & 1 \\ 1 & \frac{1}{2} - j\frac{\sqrt{3}}{2} & -\frac{1}{2} - j\frac{\sqrt{3}}{2} & -1 & \frac{1}{2} + j\frac{\sqrt{3}}{2} & \frac{1}{2} + j\frac{\sqrt{3}}{2} \\ 1 & \frac{1}{2} - j\frac{\sqrt{3}}{2} & -\frac{1}{2} + j\frac{\sqrt{3}}{2} & 1 & -\frac{1}{2} - j\frac{\sqrt{3}}{2} & -\frac{1}{2} + j\frac{\sqrt{3}}{2} \\ 1 & -1 & 1 & -1 & 1 & -1 \\ 1 & -\frac{1}{2} + j\frac{\sqrt{3}}{2} & -\frac{1}{2} - j\frac{\sqrt{3}}{2} & 1 & \frac{1}{2} + j\frac{\sqrt{3}}{2} & -\frac{1}{2} - j\frac{\sqrt{3}}{2} \\ 1 & \frac{1}{2} + j\frac{\sqrt{3}}{2} & -\frac{1}{2} + j\frac{\sqrt{3}}{2} & -1 & -\frac{1}{2} - j\frac{\sqrt{3}}{2} & \frac{1}{2} - j\frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$X_1(K) = \begin{bmatrix} 1 + 2 + 1 \\ 1 + 1 - j1.732 - 0.5 - j0.866 \\ 1 - 1 - j1.732 - 0.5 + j0.866 \\ 1 - 2 + 1 \\ 1 - \cancel{0.5} + j1.732 - 0.5 - j0.866 \\ 1 + 1 + j1.732 - 0.5 + j0.866 \end{bmatrix} = \begin{bmatrix} 4 \\ 1.5 - j2.598 \\ -0.5 - j0.866 \\ 0 \\ j0.866 - 0.5 \\ 1.5 + j2.598 \end{bmatrix}$$

$$x_2(k) = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & \frac{1}{2} - j\frac{\sqrt{3}}{2} & -\frac{1}{2} - j\frac{\sqrt{3}}{2} & -1 & -\frac{1}{2} + j\frac{\sqrt{3}}{2} & \frac{1}{2} + j\frac{\sqrt{3}}{2} \\ 1 & -\frac{1}{2} - j\frac{\sqrt{3}}{2} & -\frac{1}{2} + j\frac{\sqrt{3}}{2} & 1 & -\frac{1}{2} - j\frac{\sqrt{3}}{2} & \frac{1}{2} + j\frac{\sqrt{3}}{2} \\ 1 & -1 & 1 & -1 & 1 & -1 \\ 1 & -\frac{1}{2} + j\frac{\sqrt{3}}{2} & -\frac{1}{2} - j\frac{\sqrt{3}}{2} & 1 & -\frac{1}{2} + j\frac{\sqrt{3}}{2} & \frac{1}{2} - j\frac{\sqrt{3}}{2} \\ 1 & \frac{1}{2} + j\frac{\sqrt{3}}{2} & -\frac{1}{2} + j\frac{\sqrt{3}}{2} & -1 & -\frac{1}{2} - j\frac{\sqrt{3}}{2} & \frac{1}{2} - j\frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} 2 \\ -1 \\ 3 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$x_2(k) = \begin{bmatrix} 2 + 1 + 3 \\ 2 - 0.5 + j0.866 - 1.5 - j2.598 \\ 2 + 0.5 + j0.866 - 1.5 + j2.598 \\ 2 + 1 + 3 \\ 2 + 0.5 - j0.866 - 1.5 - j2.598 \\ 2 - 0.5 - j0.866 - 1.5 + j2.598 \end{bmatrix} = \begin{bmatrix} 4 \\ -1.732j \\ 1 + 3.464j \\ 6 \\ 1 - 3.464j \\ 1.732j \end{bmatrix}$$

$$y(k) = x_1(k) x_2(k)$$

$$= \begin{bmatrix} 4 \\ 1.5 - j2.598 \\ -0.5 - j0.866 \\ 0 \\ j0.866 - 0.5 \\ 1.5 + j2.598 \end{bmatrix} \begin{bmatrix} 2 \\ -1.732j \\ 1 + 3.464j \\ 6 \\ 1 - 3.464j \\ 1.732j \end{bmatrix} = \begin{bmatrix} 18 \\ (1.5 - j2.598)(-j1.732) \\ (-0.5 - j0.866)(1 + 3.464j) \\ 0 \\ -0.5 + j0.866(1 - 3.464j) \\ 1.732j(1.5 + j2.598) \end{bmatrix}$$

$$y(k) = \begin{bmatrix} 8 \\ -j2.598 - 4.499 \\ -0.5 - j1.732 - j0.866 + 2.999 \\ 0 \\ j0.866 + 2.999 \\ j2.598 - 4.499 \end{bmatrix} = \begin{bmatrix} 18 \\ -4.499 - j2.598 \\ 2.499 - j2.598 \\ 0 \\ 2.999 + j0.866 \\ -4.499 + j2.598 \end{bmatrix}$$

3DPT

$$y(n) = \frac{1}{6} \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & \frac{1}{2} + j\frac{\sqrt{3}}{2} & \frac{-1}{2} + j\frac{\sqrt{3}}{2} & -1 & \frac{-1}{2} - j\frac{\sqrt{3}}{2} & \frac{1}{2} - j\frac{\sqrt{3}}{2} \\ 1 & \frac{-1}{2} + j\frac{\sqrt{3}}{2} & \frac{-1}{2} - j\frac{\sqrt{3}}{2} & 1 & \frac{-1}{2} + j\frac{\sqrt{3}}{2} & \frac{-1}{2} - j\frac{\sqrt{3}}{2} \\ 1 & -1 & 1 & -1 & 1 & -1 \\ 1 & \frac{-1}{2} - j\frac{\sqrt{3}}{2} & \frac{-1}{2} + j\frac{\sqrt{3}}{2} & 1 & \frac{-1}{2} - j\frac{\sqrt{3}}{2} & \frac{-1}{2} + j\frac{\sqrt{3}}{2} \\ 1 & \frac{1}{2} - j\frac{\sqrt{3}}{2} & \frac{-1}{2} - j\frac{\sqrt{3}}{2} & -1 & \frac{1}{2} + j\frac{\sqrt{3}}{2} & \frac{1}{2} - j\frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} 16 \\ -4.5 - j2.598 \\ 2.5 - j2.598 \\ 0 \\ 2.5 + j2.598 \\ -4.5 + j2.598 \end{bmatrix}$$

$$y(n) = \frac{1}{6} \begin{bmatrix} 8 - 4.499 - j2.598 + 2.499 - j2.598 + 2.499 + j0.866 - 4.499 + j2.598 \\ 16 - 4.5 - j2.598 + 2.5 - j2.598 + 2.5 + j2.598 - 4.5 + j2.598 \\ -4.5 - j2.598 + 16 - [2.5 + j2.598](0.5 + j0.866) + 16(2.5 - j2.598)(-0.5 + j0.866) + 0 + (2.5 + j2.598)(-0.5 - j0.866) + (-4.5 + j2.598)(0.5 - j0.866) \\ 16 - (4.5 + j2.598)(-0.5 + j0.866) + (2.5 - j2.598)(-0.5 - j0.866) + (2.5 + j2.598)(-0.5 + j0.866) + (-4.5 + j2.598)(0.5 - j0.866) \\ 0 \\ 16 + 4.5 + j2.598 + 2.5 - j2.598 + 0 + 2.5 + j2.598 + 4.5 - j2.598 \\ 16 - (4.5 + j2.598)(-0.5 - j0.866) + (2.5 - j2.598)(-0.5 + j0.866) + (2.5 + j2.598)(-0.5 - j0.866) + (-4.5 + j2.598)(0.5 + j0.866) \\ 16 - (4.5 + j2.598)(0.5 - j0.866) + (2.5 - j2.598)(-0.5 - j0.866) + 0 \\ (2.5 + j2.598)(-0.5 + j0.866) + (-4.5 + j2.598)(0.5 + j0.866) \end{bmatrix}$$

$$y(n) = \frac{1}{6} \begin{bmatrix} 12 \\ 18 - 99 = 18 \\ 11.9 = 12 \\ 30 \\ 18 \\ 5.28 \times 10^{-4} \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ 2 \\ 5 \\ 3 \\ 0 \end{bmatrix}$$

$$\therefore y(n) = \{2, 3, 2, 5, 3, 0\}$$

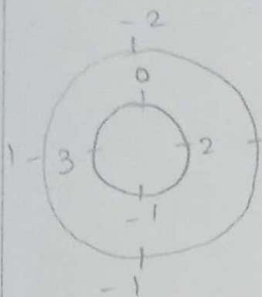
3. Perform circular convolution bet the following discrete sequences using graphical.

$$x(n) = \{1, -2, 1, -1\}, h(n) = \{2, -1, 3\}$$

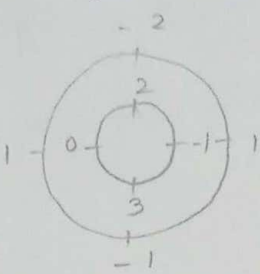
Soln

$$N_1 = 4, N_2 = 3$$

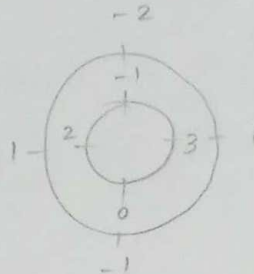
$$\therefore h(n) = \{2, -1, 3, 0\}$$



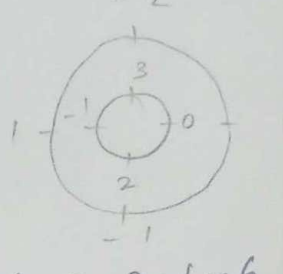
$$y(0) = 2 + 1 + 3 + 0 = 6$$



$$y(1) = -1 - 3 + 0 - 4 = -8$$



$$y(2) = 3 + 0 + 2 + 2 = 7$$



$$y(3) = 0 - 2 - 1 - 6 = -9$$

$$y(n) = \{6, -8, 7, -9\}$$

circular convolution

$$x(n) = \{1, 2, 3\}$$

$$h(n) = \{-1, 2\}$$

$$N = 3$$

add one zero to $h(n)$

$$\therefore h(n) = \{-1, 2, 0\}$$

using matrix method.

LC using CC

$$x(n) = \{1, 2, 3\}$$

$$h(n) = \{-1, 2\}$$

$$N_1 = 3, N_2 = 2$$

$$N = 3 + 2 - 1 = 4$$

$$\therefore x(n) = \{1, 2, 3, 0\}$$

$$h(n) = \{-1, 2, 0, 0\}$$

9/7/26

Linear filtering approach - using FFT
 $\rightarrow$ finding the response

(L.F)

$$y(n) = x(n) * h(n) = \sum_{k=-\infty}^{\infty} x(n) h(n-k)$$

$\rightarrow$ filter coefficient

$\rightarrow$ requires more time

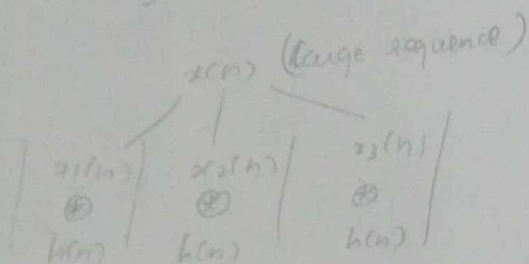
$\rightarrow$ requires more memory

$\rightarrow$ when N is very large

That's why we use FFT to compute N -pt DFT with more speed.

FFT. (Basic principle)

↓
Decomposition of i/p response
(dividing/splitting) the longest sequence into small sequence



overlap & add method (OLA)

overlap & save method (OLS)

Overlap & Add method :

$$x(n) = \{-1, 2, 1, -1, 2, 3\}$$

$$h(n) = \{1, -1, 1\}$$

1. $M \rightarrow$ length of filter coefficient $= 3$

2. $L \rightarrow$ length of non-overlapping block

Obtain from longest sequence $= 3$ | or divide in length of filter

$$x(n) = \{-1, 2, 1 \mid -1, 2, 3\}$$

$$x_1(n) = \{-1, 2, 1\}$$

$$x_2(n) = \{-1, 2, 3\}$$

length of FFT

$$N \geq L + M - 1$$

$$N = 3 + 3 - 1$$

$$N = 5$$

choose as well greater than 5 not less than 5

↓
also length of individual response also be equal to N .

4. pad required no. of zeros at the end of $x_1(n)$, $x_2(n)$ & $h(n)$

$$x_1'(n) = \{ \overset{0}{-1}, \overset{0}{2}, \overset{0}{1}, \overset{0}{0}, \overset{0}{0} \}$$

$$h'(n) = \{ \overset{0}{1}, \overset{0}{-1}, \overset{0}{1}, \overset{0}{0}, \overset{0}{0} \}$$

$$x_1'(n) \otimes h'(n) = y_1(n)$$

$$sp \rightarrow 0+0=0$$

$$x_2'(n) = \{ \overset{0}{-1}, \overset{0}{2}, \overset{0}{3}, \overset{0}{0}, \overset{0}{0} \}$$

$$h'(n) = \{ \overset{0}{1}, \overset{0}{-1}, \overset{0}{1}, \overset{0}{0}, \overset{0}{0} \}$$

$$x_2'(n) \otimes h'(n) = y_2(n)$$

$$sp \rightarrow 3+0=3$$

Find $y_1(n)$

$$\begin{bmatrix} -1 & 0 & 0 & 1 & 2 \\ 2 & -1 & 0 & 0 & 1 \\ 1 & 2 & -1 & 0 & 0 \\ 0 & 1 & 2 & -1 & 0 \\ 0 & 0 & 1 & 2 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} -1+0+0+0+0 \\ 2+1+0+0+0 \\ 1-2-1+0+0 \\ 0-1+2+0+0 \\ 0+0+1+0+0 \end{bmatrix} = \begin{bmatrix} -1 \\ 3 \\ -2 \\ 1 \\ 1 \end{bmatrix}$$

Find $y_2(n)$

$$\begin{bmatrix} -1 & 0 & 0 & 3 & 2 \\ 2 & -1 & 0 & 0 & 3 \\ 3 & 2 & -1 & 0 & 0 \\ 0 & 3 & 2 & -1 & 0 \\ 0 & 0 & 3 & 2 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} -1+0+0+0+0 \\ 2+1 \\ 3-2-1 \\ 0-3+2 \\ 0+0+3+0-1 \end{bmatrix} = \begin{bmatrix} -1 \\ 3 \\ 0 \\ -1 \\ 2 \end{bmatrix}$$

$$y_1(n) = \{-1, 3, -2, 1, 1\}$$

$$y_2(n) = \{-1, 3, 0, -1, 3\}$$

starting pt (sp) $\rightarrow (sp)_1 + (sp)_2$

n $\rightarrow$	0	1	2	3	4	5	6	7
$y_1(n)$	-1	3	-2	1	1			
$y_2(n)$				-1	3	0	-1	3
$y(n)$	-1	3	-2	0	4	0	-1	3

Individual response

Overall response

last (M-1) pt of 1st sequence overlapped with first (M-1) pt of second sequence.

	-1	0	1	-1	2	3
1	-1	2	1	-1	2	3
-1	1	-2	-1	1	-2	-3
1	-1	2	1	-1	2	3

-1	2	1	-1	2	3
1	-1	1			

$$y(n) = \{-1, 3, -2, 0, 4, 0, -1, 3\}$$

shortest using
Tabular method
is worthy.

2. $x(n) = \{-3, 4, 1, 0, 2, -5, 7\}$ determine the response
 $h(n) = \{4, 6, 2\}$ of the LTI system with an

$a(n)$ & impulse response $h(n)$ using overlap and
add method.

Soln

i) $M=3$

ii) $L=4$

iii) $N \geq L+M-1$

$N \geq 4+3-1$

$N \geq 6$

iv) $x_1(n) = \{-3, 4, 1, 0\}$

$x_2(n) = \{2, -5, 7, 0\}$

if sequence is periodic
 $\{2, -5, 7, -3\}$

	-3	4	1	0	2	-5	7
4	-12	16	4	0	8	-25	28
6	-18	24	6	0	12	-30	42
2	-6	8	2	0	4	-10	14

-12, -2, 22, 14, 10, -18, 2, 32

After padding.

$x_1'(n) = \{-3, 4, 1, 0, 0, 0\}$, $h_1'(n) = \{4, 6, 2, 0, 0, 0\}$

$x_2'(n) = \{2, -5, 7, 0, 0, 0\}$

$y_1(n) = x_1'(n) \otimes h_1'(n)$

$$\begin{bmatrix} -3 & 0 & 0 & 0 & 1 & 4 \\ 4 & -3 & 0 & 0 & 0 & 1 \\ 1 & 4 & -3 & 0 & 0 & 0 \\ 0 & 1 & 4 & -3 & 0 & 0 \\ 0 & 0 & 1 & 4 & -3 & 0 \\ 0 & 0 & 0 & 1 & 4 & -3 \end{bmatrix} \begin{bmatrix} 4 \\ 6 \\ 2 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} -12+0 \\ 16-18 \\ 4+24-6 \\ 6+8 \\ 2 \\ 0 \end{bmatrix} = \begin{bmatrix} -12 \\ -2 \\ 22 \\ 14 \\ 2 \\ 0 \end{bmatrix}$$

$$y_2(n) = x_1(n) \otimes h_1(n)$$

$$\begin{bmatrix} 2 & 0 & 0 & 0 & 7 & -5 \\ -5 & 2 & 0 & 0 & 0 & 7 \\ 7 & -5 & 2 & 0 & 0 & 0 \\ 0 & 7 & -5 & 2 & 0 & 0 \\ 0 & 0 & 7 & -5 & 2 & 0 \\ 0 & 0 & 0 & 7 & -5 & 2 \end{bmatrix} \begin{bmatrix} 4 \\ 6 \\ 2 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 8 \\ -20+12 \\ 28-30+4 \\ 42-10 \\ 14 \\ 0 \end{bmatrix} = \begin{bmatrix} 8 \\ -8 \\ 2 \\ 32 \\ 14 \\ 0 \end{bmatrix}$$

n	0	1	2	3	4	5	6	7	8	
$y_1(n)$	-12	-2	22	14	2	0				
$y_2(n)$					8	-8	2	32	14	0
$y(n)$	-12	-2	22	14	10	-8	2	32	14	0

length of response = 7 but $y(n) = 10$ but don't affect the length when zero is at end of the overall response.

overlap and save method ^{dividing the} longest sequence can be decimated by any overlapping.

$$x(n) = \{-3, 4, 1, 0, 2, -5, 7\} \quad \& \quad h(n) = \{4, 6, 2\}$$

1. $m = 3$

2. choose $N = 6$

$$N \geq L + m - 1$$

$$N - m + 1 \geq L$$

$$6 - 3 + 1$$

$$\boxed{4 \geq L}$$

$$\boxed{L = 4}$$

$$x_1(n) = \{-3, 4, 1, 0\} \rightarrow \text{Pad } (M-1) \text{ zeros at beginning of 1st sequence}$$

$$x_2(n) = \{2, -5, 7, 0\} \rightarrow \text{append last } (M-1) \text{ elements of 1st sequence to beginning of 2nd sequence}$$

$$\rightarrow x_1'(n) = \{0, 0, -3, 4, 1, 0\}$$

$$x_2'(n) = \{1, 0, 2, -5, 7, 0\}$$

$$y_1(n) = x_1'(n) * h'(n)$$

$$h'(n) = \{4, 6, 2, 0, 0, 0\}$$

$y_1(n)$

$$\begin{bmatrix} 0 & 0 & 1 & 4 & -3 & 0 \\ 0 & 0 & 0 & 1 & 4 & -3 \\ -3 & 0 & 0 & 0 & 1 & 4 \\ 4 & -3 & 0 & 0 & 0 & 1 \\ 1 & 4 & -3 & 0 & 0 & 0 \\ 0 & 1 & 4 & -3 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 6 \\ 2 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ -12 \\ 16 + (-18) \\ 4 + 24 - 6 \\ 6 + 8 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ -12 \\ -2 \\ 22 \\ 14 \end{bmatrix}$$

$y_2(n)$

$$\begin{bmatrix} 1 & 0 & 4 & -5 & 2 & 0 \\ 0 & 1 & 0 & 7 & -5 & 2 \\ 2 & 0 & 1 & 0 & 7 & -5 \\ -5 & 2 & 0 & 1 & 0 & 7 \\ 7 & -5 & 2 & 0 & 1 & 0 \\ 0 & 7 & -5 & 2 & 0 & 1 \end{bmatrix} \begin{bmatrix} 4 \\ 6 \\ 2 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 4 + 14 \\ 6 \\ 8 + 2 \\ -20 + 12 \\ 28 - 30 + 4 \\ 42 - 10 \end{bmatrix} = \begin{bmatrix} 18 \\ 6 \\ 10 \\ -8 \\ 2 \\ 32 \end{bmatrix}$$

remove the first (m-1) elements of 1st & 2nd sequence (each sequence)

n	0	1	2	3	4	5	6	7	8	9
$y_1(n)$	2	0	-12	-2	22	14				
$y_2(n)$				18	6	10	-8	2	32	
$y(n)$			-12	-2	22	10	10	-8	2	32

2nd/6th shift overlap add result

13/7/26

FFT Algorithms Basic principle: Decimation

FFT:

It can be used to determine N-pt (DFT) with more speed

FFT algorithm:

DIT

DIF

When the N-length of the sequence is represented as 2^L (L is applicable to 2)
called radix 2. Eg. N=8

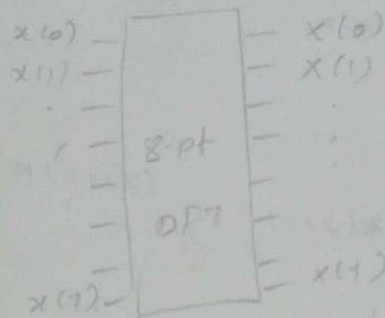
11/11/2019 radix 2

$$N = 2 \cdot 2 \cdot 2 = 2^3 \rightarrow \text{radix-2}$$

Radix-2

$N = 2^L \rightarrow$ no. of stages in DFT.
base

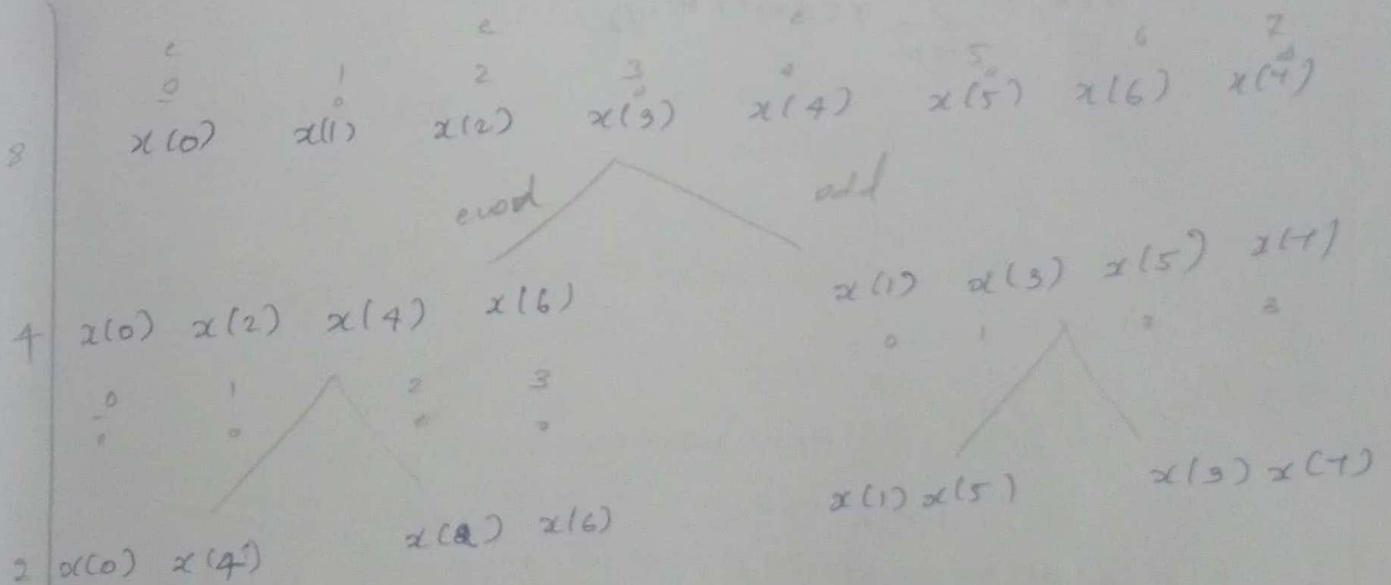
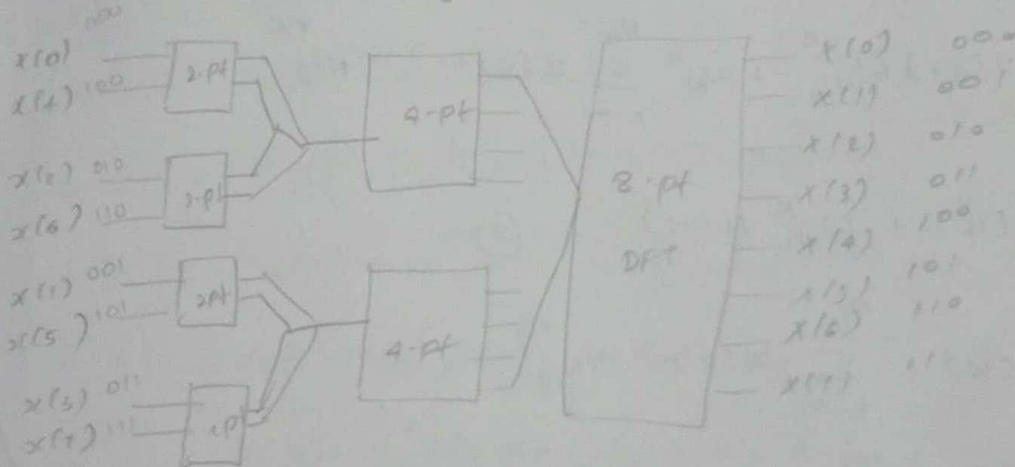
In radix-2 DFT algorithm input is bit reversed & output is normal manner



after decimation

bit reversed manner

ifp is shuffled after decimation



No. of stages involved in FFT using decimation algorithm is

$$L = \log_2 N$$

$$L = \log_2 N$$

Derivation of Radix-2 DIT-FFT algorithm.

$$N\text{-pt DFT } [x(n)] = X(K) = \sum_{n=0}^{N-1} x(n) e^{-j\frac{2\pi}{N}nk}$$

$$X(K) = \sum_{n=0}^{N-1} x(n) W_N^{nK} \rightarrow (1)$$

Let $n=2r$ for even

$n=2r+1$ for odd.

$$\begin{aligned} X(K) &= \sum_{r=0}^{\frac{N}{2}-1} x(2r) W_N^{2rk} + \sum_{r=0}^{\frac{N}{2}-1} x(2r+1) W_N^{(2r+1)K} \\ &= \sum_{r=0}^{\frac{N}{2}-1} x(2r) W_{N/2}^{rk} + \sum_{r=0}^{\frac{N}{2}-1} x(2r+1) W_{N/2}^{rk} \cdot W_N^{K} \\ &= \underbrace{\sum_{r=0}^{\frac{N}{2}-1} x(2r) W_{N/2}^{rk}}_{N/2 \text{ pt DFT}} + W_N^K \underbrace{\sum_{r=0}^{\frac{N}{2}-1} x(2r+1) W_{N/2}^{rk}}_{N/2 \text{ pt DFT}} \end{aligned}$$

$$X(K) = G(K) + W_N^K H(K) \rightarrow (2)$$

where,

$$G(K) = \sum_{r=0}^{\frac{N}{2}-1} x(2r) W_{N/2}^{rk}$$

$$H(K) = \sum_{r=0}^{\frac{N}{2}-1} x(2r+1) W_{N/2}^{rk}$$

$$X(K) = G(K) + W_N^K H(K)$$

$$X(0) = G(0) + W_8^0 H(0)$$

$$X(1) = G(1) + W_8^1 H(1)$$

$$X(2) = G(2) + W_8^2 H(2)$$

$$X(3) = G(3) + W_8^3 H(3)$$

$$X(4) = G(4) + W_8^4 H(4)$$

$$X(5) = G(5) + W_8^5 H(5)$$

$$X(6) = G(6) + W_8^6 H(6)$$

$$X(7) = G(7) + W_8^7 H(7)$$

Hence $G(K)$ and $H(K)$ is a p)

we need to 8pt

Assume $G(K)$ & $H(K)$ is periodic with the period $N/2$. ($N=8$)

$$G(K) = G(K + N/2)$$

$$H(K) = H(K + N/2)$$

$$\begin{aligned} G(0) &= G(4) \\ G(1) &= G(5) \\ G(2) &= G(6) \\ G(3) &= G(7) \\ H(0) &= H(4) \\ H(1) &= H(5) \\ H(2) &= H(6) \\ H(3) &= H(7) \end{aligned}$$

$$\begin{aligned} G(0) &= G(0/2) = G(0) \\ H(0) &= H(0/2) = H(0) \end{aligned}$$

$$X(0) = G(0) + W_8^0 H(0)$$

$$X(1) = G(1) + W_8^1 H(1)$$

$$X(2) = G(2) + W_8^2 H(2)$$

$$X(3) = G(3) + W_8^3 H(3)$$

$$X(4) = G(0) + W_8^4 H(0) = G(0) - W_8^0 H(0)$$

$$X(5) = G(1) + W_8^5 H(1) = G(1) - W_8^1 H(1)$$

$$X(6) = G(2) + W_8^6 H(2) = G(2) - W_8^2 H(2)$$

$$X(7) = G(3) + W_8^7 H(3) = G(3) - W_8^3 H(3)$$

$$W_8^0 = 1$$

$$W_8^1 = e^{-j\frac{2\pi}{8}}$$

$$W_8^2 = e^{-j\frac{2\pi}{4}} = -1$$

$$W_8^3 = e^{-j\frac{2\pi}{8} \cdot 3} = -j$$

$$W_8^4 = e^{-j\frac{2\pi}{2}} = -1$$

$$W_8^5 = e^{-j\frac{2\pi}{8} \cdot 5} = j$$

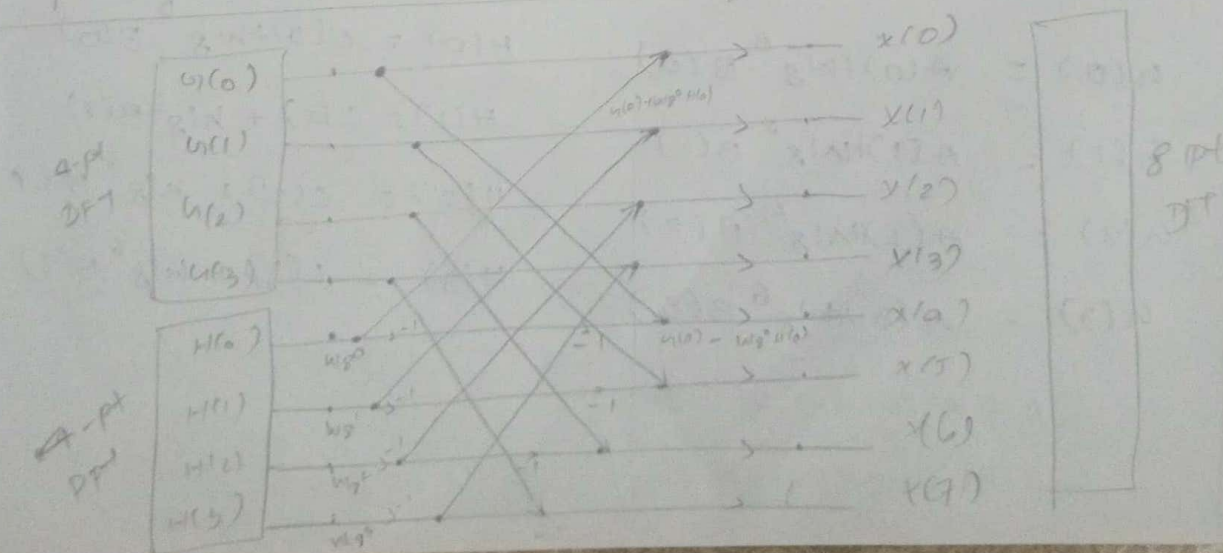
$$W_8^6 = e^{-j\frac{2\pi}{4} \cdot 3} = 1$$

$$W_8^7 = e^{-j\frac{2\pi}{8} \cdot 7} = -j$$

To compute 2-pt DFT, 2-Addition
1-multiplication $\times 4$

therefore to compute 8-pt DFT need 8 addition and 4 multiplication.

compare to Direct method we need $N(N-1)$ add & N^2 mult



II stage

$h(k) \rightarrow N/2$ DFT

$$h(k) = \sum_{r=0}^{N/2-1} g(r) w_{N/2}^{rk}$$

$$g(r) \begin{cases} \text{even} : r = 2l \\ \text{odd} : r = 2l+1 \end{cases}$$

$$h(k) = \sum_{l=0}^{N/4-1} g(2l) w_{N/2}^{2lk} + \sum_{l=0}^{N/4-1} g(2l+1) w_{N/2}^{(2l+1)k}$$

$$= \sum_{l=0}^{N/4-1} g(2l) w_{N/4}^{lk} + w_{N/2}^k \sum_{l=0}^{N/4-1} g(2l+1) w_{N/4}^{lk}$$

$$h(k) = \sum_{l=0}^{N/4-1} g(2l) w_{N/4}^{lk} + w_N^{2k} \sum_{l=0}^{N/4-1} g(2l+1) w_{N/4}^{lk}$$

$\frac{N}{2}$ pt DFT

$\frac{N}{4}$ pt DFT

$\frac{N}{4}$ pt DFT

$$h(k) = A(k) + w_N^{2k} B(k)$$

$$H(k) = C(k) + w_N^{2k} D(k)$$

$$h(0) = A(0) + w_8^0 B(0)$$

$$h(1) = A(1) + w_8^2 B(1)$$

$$h(2) = A(2) + w_8^4 B(2)$$

$$h(3) = A(3) + w_8^6 B(3)$$

$$H(0) = C(0) + w_8^0 B(0)$$

$$H(1) = C(1) + w_8^2 B(1)$$

$$H(2) = C(2) + w_8^4 B(2)$$

$$H(3) = C(3) + w_8^6 B(3)$$

A & B are 2 pt DFT loc $\frac{N}{4} = \frac{8}{4} = 2$

but here have 4 pt DFT so apply recursively prop

$$A(k) \text{ or } B(k) = A(k + N/4) + B(k + N/4)$$

$$H(k) = C(k + N/4) + B(k + N/4)$$

$$u(0) = A(0) + w_8^0 B(0)$$

$$u(0) = A(0) + B(0)$$

$$u(1) = A(1) + w_8^2 B(1)$$

$$u(1) = A(2) + B(3)$$

$$u(3) = A(0) + w_8^4 B(0) = A(0) - w_8^0 B(0)$$

$$u(4) = A(1) + w_8^6 B(1) = A(1) - w_8^2 B(1)$$

$$H(0) = C(0) + w_8^0 D(0)$$

$$H(1) = C(1) + w_8^2 D(1)$$

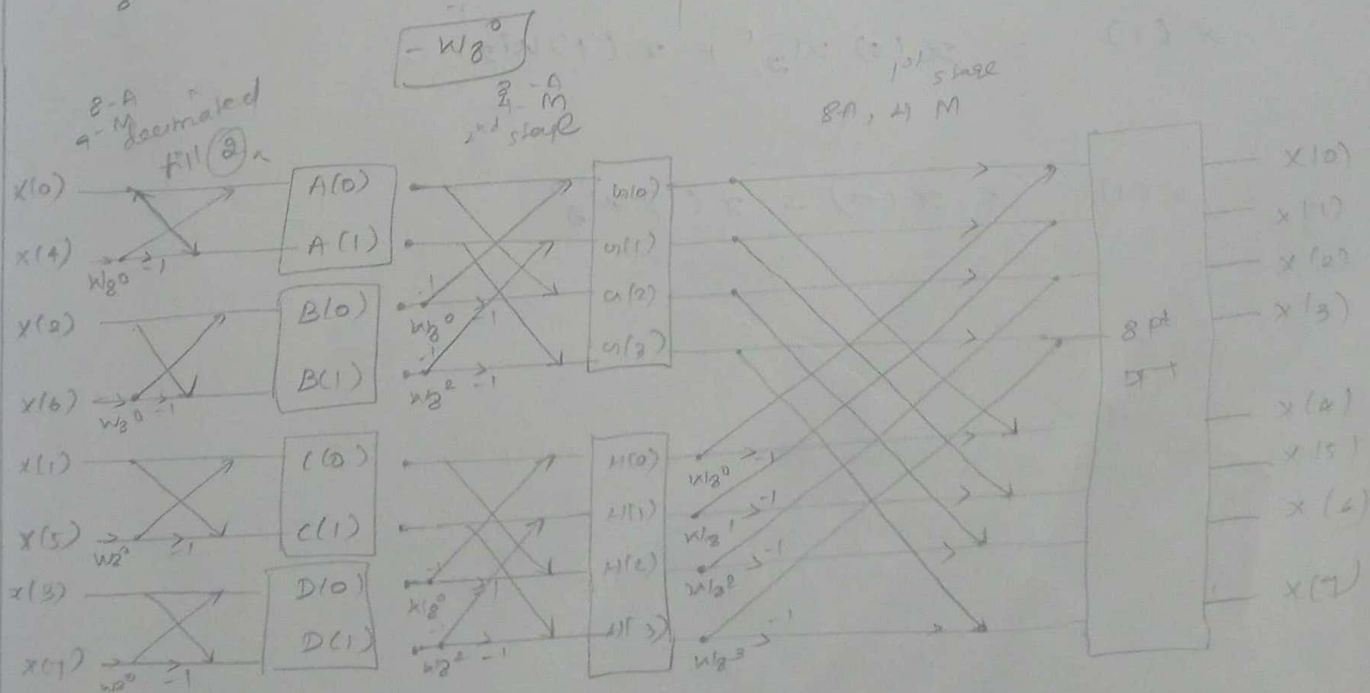
$$H(3) = C(0) + w_8^4 D(0) = C(0) - w_8^0 D(0)$$

$$H(4) = C(1) + w_8^6 D(1) = C(1) - w_8^2 D(1)$$

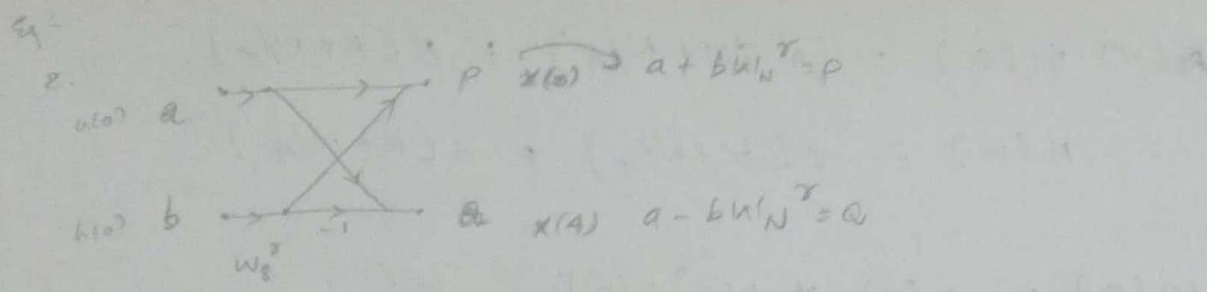
$$w_N^k = w_N^{k+N/2}$$

$$w_8^6 = w_8^{2+8/2} = -w_8^2$$

$$w_8^4 = e^{-j2\pi \cdot 4/8} = e^{-j\pi} = -1$$



In each stage bottom half stage is multiplied with twiddle factor



(common general structure) in all stages.

It is Basic butterfly structure used in radix-2 DIT-FFT algorithm.

$$x(k) = \sum_{n=0}^{N-1} x(n) w_N^{nk}$$

$N=2$

$$x(k) = \sum_{n=0}^1 x(n) w_2^{nk}$$

in order to all middle factors in common both 2.

$$x(0) = x(0) w_2^0 + x(1) w_2^0 \rightarrow a + b w_N^0$$

$$x(0) = x(0) + x(1) w_2^0$$

compute by general structure

$$x(1) = x(0) w_2^0 + x(1) w_2^1 \rightarrow a - b w_N^0$$

$-w_2^1 \rightarrow -w_2^0$

$$x(1) = x(0) - x(1) w_2^0$$

bottom factor

- 64 rows
- Buffer
- Complex complexity (84 A & 12 M) compared to
- in direct method 64 mult & 56 addition
- so % of complex is reduced.

$$M = \frac{N}{2} \log_2 N$$

$$A = N \log_2 N$$

stages: $\log_2 N$

Draw the butterfly structure to compute 4 pt DFT using radix-2 decimation in time (DIT) FFT algorithm

step 1 $\rightarrow$ find the no. of stages

step 2 $\rightarrow$ draw structure and no. of lines

step 3 $\rightarrow$ determine input sequences. (based on even/odd)

step 4 $\rightarrow$ start to draw butterfly diagram from a 2 pt DFT stage

step 5 $\rightarrow$ Place bottom of stage as twiddle factor $w_N^k = w_N^{2k}$ $k=0,1,2,\dots$

step 6 $\rightarrow$ which has twiddle factor can multiply with (-1) and combine two inputs in 1st stage (rept)

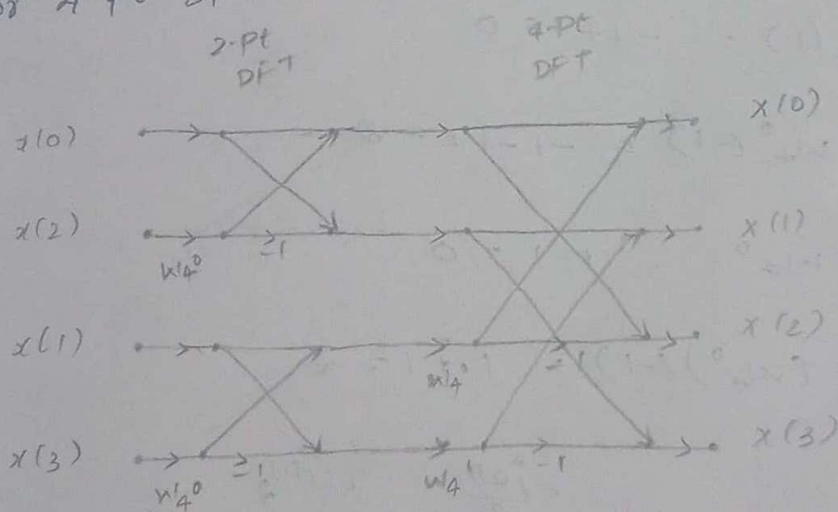
step 7 $\rightarrow$ repeat (from ^{step 5} other stages

first stage always Twiddle factor is w_N^0

and last stage always, if 8-pt, $w_N^0 - w_N^1$ if 4 $w_N^0 - w_N^2$

middle stage $w_N^{N/4}$

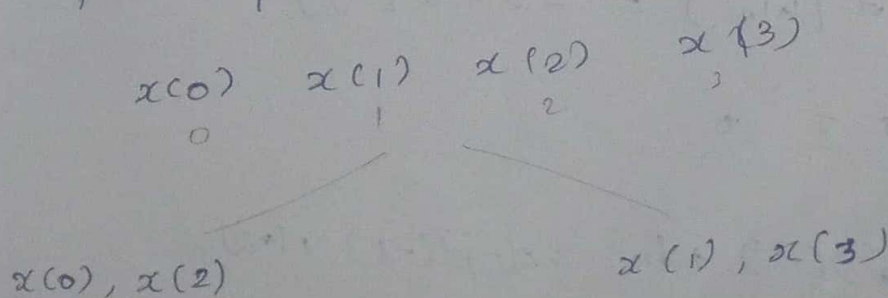
For 4-pt DFT:



(i) $N=4$

(ii) stages: $= \log_2 N = \log_2 4 = 2$

(iii) Input sequences



compute 'N'-pt DFT of the given sequences using DIT-FFT algorithm.

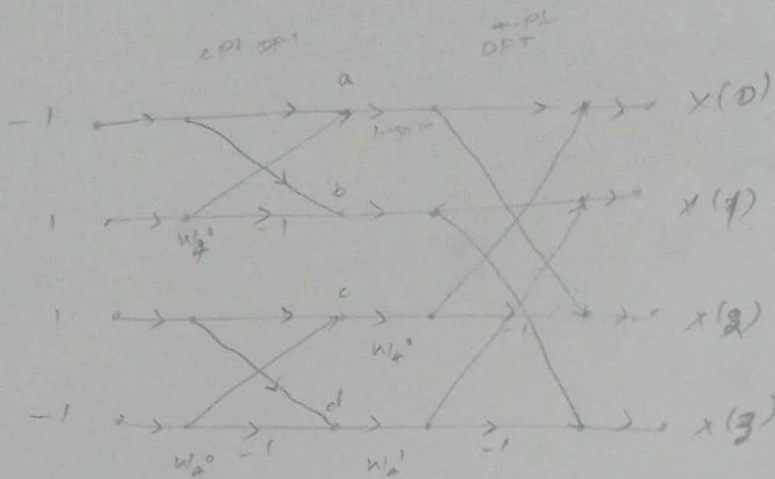
1. $x(n) = \{ -1, 1, 1, -1 \}$.

$x(0), x(2)$

$-1, 1$

$x(1), x(3)$

$1, -1$



I-stage

$$a = -1 + w_4^0(1) = -1 + 1 = 0$$

$$b = -1 + (1) w_4^0(-1) = -1 - 1 = -2$$

$$c = 1 + (-1) w_4^0 = 1 - 1 = 0$$

$$d = 1 + (-1) (w_4^0)(-1) = 1 + 1 = 2$$

here, $w_4^0 = 1$, $w_4^1 = e^{-\frac{j2\pi}{4}} = e^{-j\pi/2} = -j$.

II-stage

$$x(0) = a + c[w_4^0] = 0 + 0(1) = 0$$

$$x(1) = b + d[w_4^1] = -2 + 2[-j] = -2 - 2j$$

$$x(2) = -(c) + a = 0$$

$$x(3) = d(w_4^1)(-1) + b = 2(-j)(-1) + (-2) = -2 + 2j$$

$$x(k) = \{0, -2-2j, 0, -2+2j\}$$

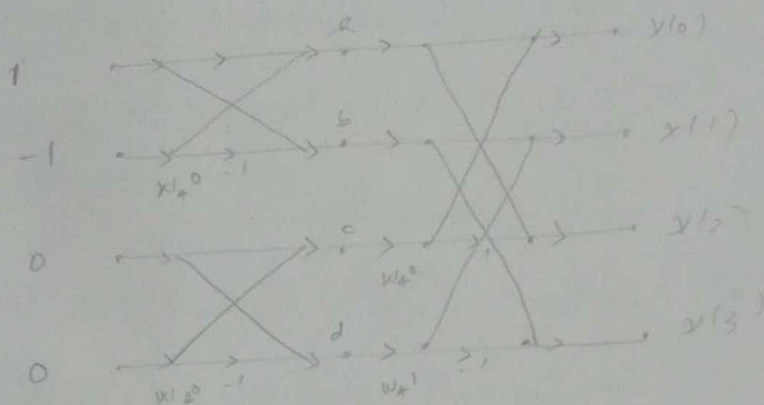
$$\cos\left(\frac{n\pi}{2}\right) = x(n)$$

$$x(0) = 1$$

$$x(1) = 0$$

$$x(2) = -1$$

$$x(3) = 0$$



I-stage

$$a = 1 + (-1)w_4^0 = 1 - 1 = 0$$

$$b = 1 + (-1)(w_4^0)(-1) = 1 + (-1)(-1) = 1 + 1 = 2$$

$$c = 0 + 0 = 0$$

$$d = 0 + 0 = 0$$

Handwritten notes:
 $w_4^0 = 1$
 $w_4^1 = j$

II-stage

$$x(0) = a + c(w_4^0)(-1) = 0 + (0) = 0$$

$$x(1) = b + d(w_4^1)(-1) = 2 + 0 = 2$$

$$x(2) = c(w_4^0)(-1) + a = 0 + 0 = 0$$

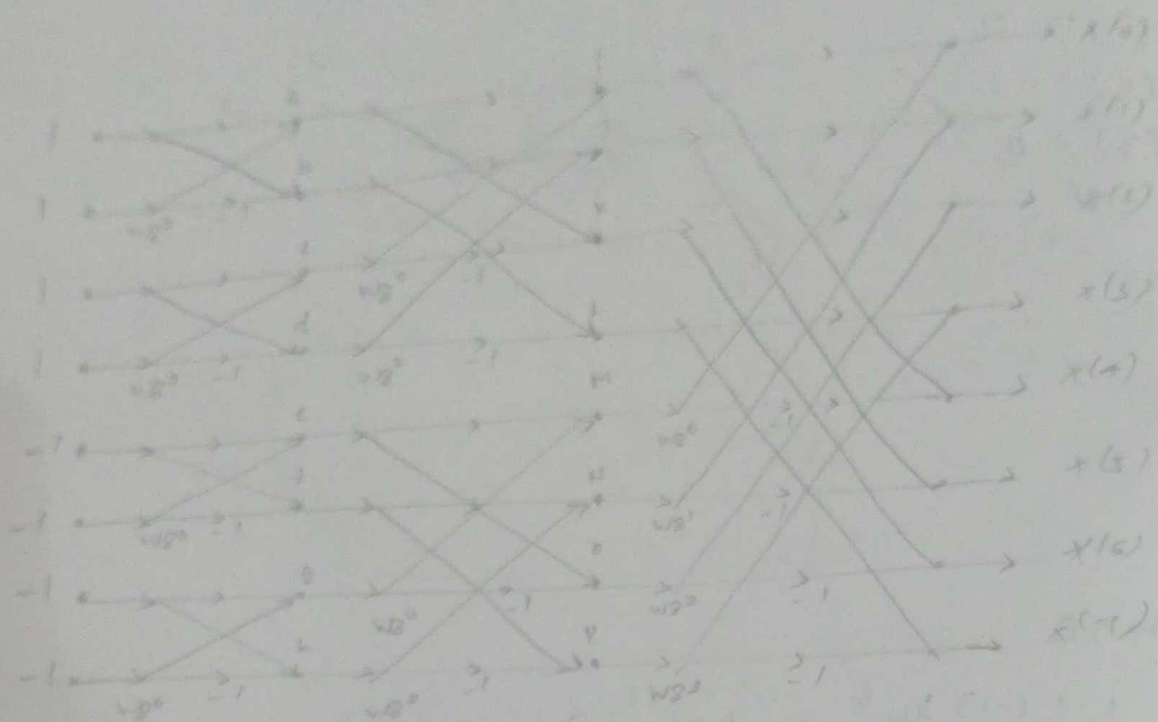
$$x(3) = d(w_4^1)(-1) + b = 0 + 2$$

$$x(k) = \{0, 2, 0, 2\}$$

compute N-pt DFT of the given sequence using radix-2

$$x(n) = \{1, -1, 1, -1, 1, -1, 1, -1\}$$

DIT FFT algorithm.



I stage

$$a = 1 + 1(w_8^0) = 1 + 1 = 2$$

$$b = 1 + 1(w_8^0)(-1) = 1 - 1 = 0$$

$$c = 1 + 1(w_8^0) = 1 + 1 = 2$$

$$d = 1 + 1(w_8^0)(-1) = 1 - 1 = 0$$

$$e = -1 + (-1)(w_8^0) = -1 - 1 = -2$$

$$f = -1 + (-1)w_8^0(-1) = -1 + 1 = 0$$

$$g = -1 + (-1)w_8^0 = -2$$

$$h = -1 + (-1)w_8^0(-1) = -1 + 1 = 0$$

$$w_8^2 = e^{-\frac{j2\pi}{8} \cdot 2} = e^{-j\pi/2} = -j$$

$$w_8^4 = e^{-\frac{j2\pi}{8} \cdot 4} = e^{-j\pi} = -1$$

$$w_8^6 = e^{-\frac{j2\pi}{8} \cdot 6} = e^{-j3\pi/2} = j$$

II-stage

$$i = a + c(w_8^0) = 2 + 2(1) = 4$$

$$j = b + d(w_8^2) = 0 + 0 = 0$$

$$k = a + c(w_8^0)(-1) = 2 + 2(1)(-1) = 2 - 2 = 0$$

$$l = b + d(w_8^2)(-1) = 0 + 0 = 0$$

$$m = e + g(w_8^0) = -2 + (-2)(1) = -4$$

$$n = f + h(w_8^2) = 0 + 0 = 0$$

$$o = e + g(w_8^0)(-1) = -2 + (-2)(1)(-1) = -2 + 2 = 0$$

$$p = f + h(w_8^2)(-1) = 0 + 0 = 0$$

III-stage

$$x(0) = i + m(w_8^0) = 4 + (-4)(1) = 0$$

$$x(1) = j + n(w_8^1) = 0 + 0 = 0$$

$$x(2) = k + o(w_8^2) = 0$$

$$x(3) = l + p(w_8^2) = 0 + 0 = 0$$

$$x(4) = i + m(w_8^0)(-1) = 4 + (-4)(-1)(1) = 8$$

$$x(5) = j + n(w_8^1)(-1) = 0 + 0 = 0$$

$$x(6) = k + o(w_8^2)(-1) = 0 + 0$$

$$x(7) = l + p(w_8^3)(-1) = 0$$

$$x(k) = \{0, 0, 0, 0, 8, 0, 0, 0\}$$

2. $x(n) = \{0, 1, 2, 3, 4, 5, 6, 7\}$ using

DIT-FFT algorithm

So(n)

DIF - FFT algorithm

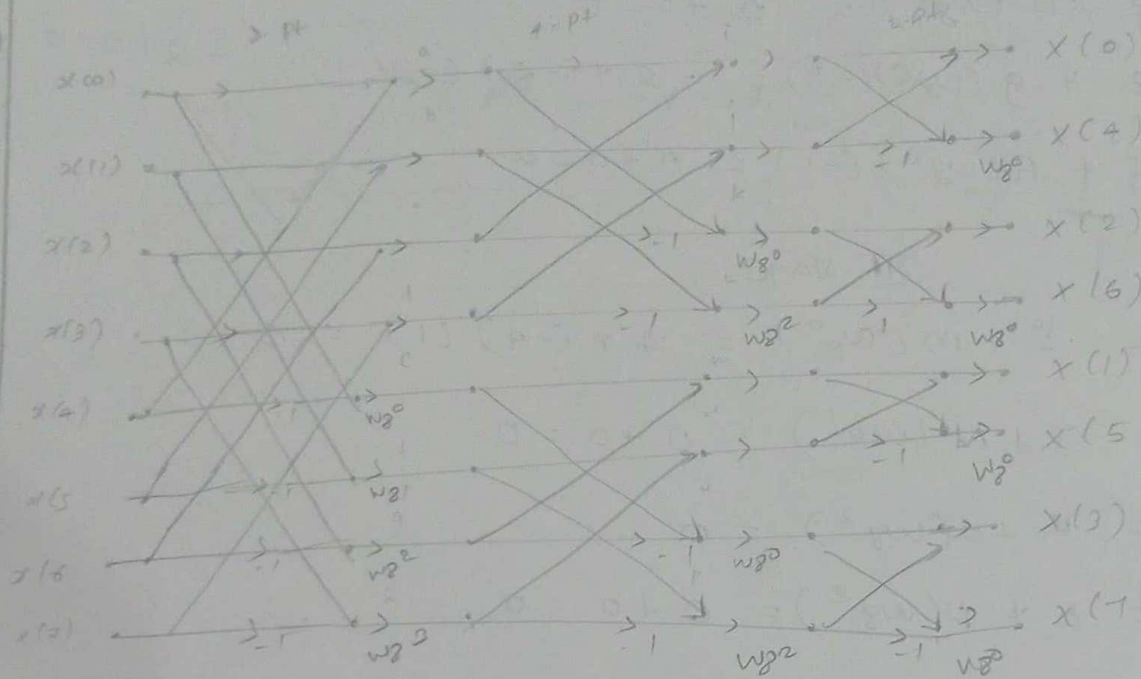
apply decimation in output.

i/p $\rightarrow$ normal order

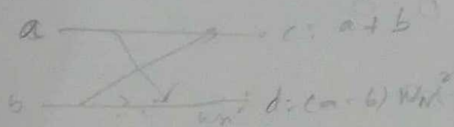
o/p $\rightarrow$ bit reversed sequence.

1. change the structure
2. ifp kept normal
o/p will change
bit reversed

7m

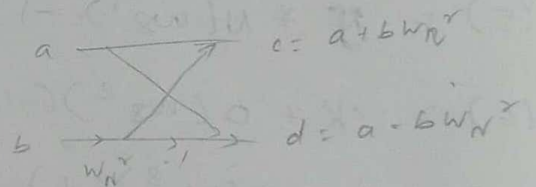


General structure of DIF



2-A & 1-M

General structure of DIT



2-A & 1-M

There is no change computational requirements for DIT & DIF

Total 2A Addition & 12 multiplications

DIT	DIF
i/p is decimated o/p is normal & i/p is bit reversed Same A & M	o/p is decimated o/p is bit reversed & i/p is normal Same A & M

Q. How many no. of multiplication and addition are required 1024 pt DFT. by using direct method and radix DIT FFT & DIF FFT method.

soln $N=1024$
Direct method

$$A = N(N-1) = 1024(1023) = 1047552$$

$$M = N^2 = (1024)^2 = 1048576$$

In Both DIT & DIF method.

$$A = N \log_2 N = 1024 \log_2 1024$$

$$= (1024) \log_2 2^{10}$$

$$= (1024) 10 \cdot 1$$

$$A = 10240$$

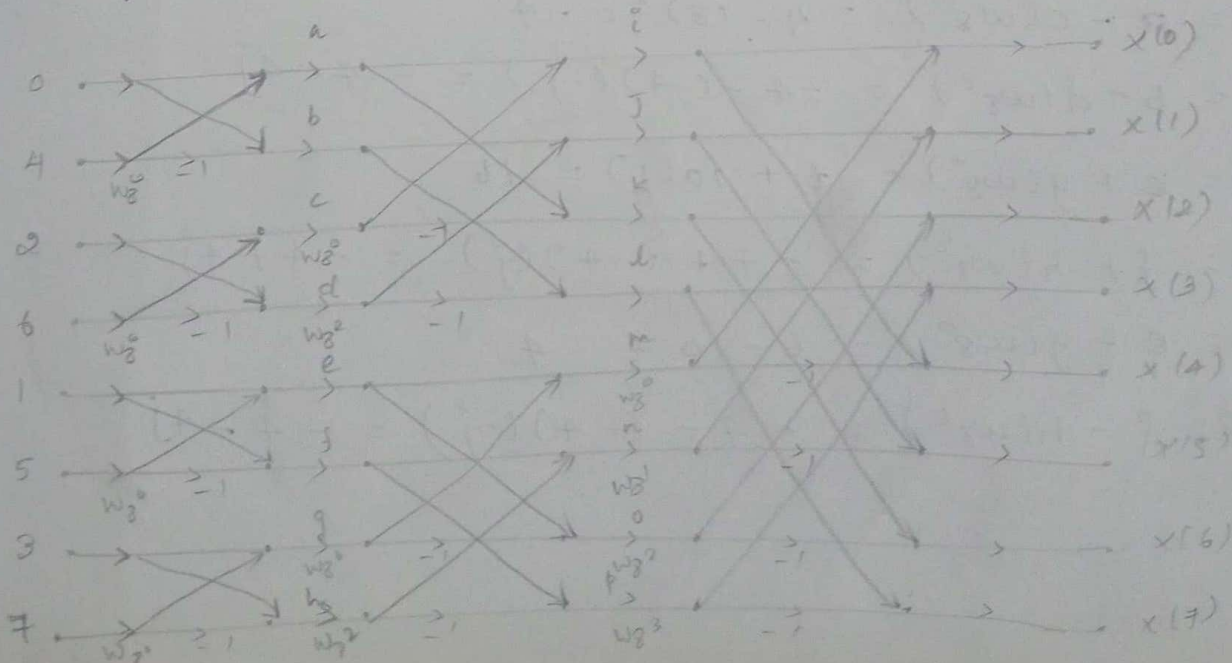
$$M = \frac{N}{2} \log_2 N = \frac{1024}{2} \log_2 1024$$

$$= 512 \times 10 \log_2 2$$

$$M = 5120$$

fw

$$x(n) = \{0, 1, 2, 3, 4, 5, 6, 7\}$$



I-stage

$$a = 0 + 4(\omega_8^0) = 4$$

$$b = 0 + 4(\omega_8^0)(-1) = 0 + 4(-1) = -4$$

$$c = 2 + 6(\omega_8^0) = 2 + 6 = 8$$

$$d = 2 - 6(\omega_8^0) = 2 - 6 = -4$$

$$e = 1 + 5(\omega_8^0) = 1 + 5 = 6$$

$$f = 1 - 5(\omega_8^0) = 1 - 5 = -4$$

$$g = 3 + 7(\omega_8^0) = 3 + 7 = 10$$

$$h = 3 - 7(\omega_8^0) = 3 - 7 = -4$$

II-stage

$$i = a + c(\omega_8^0) = 4 + 8(1) = 12$$

$$j = b + d(\omega_8^2) = -4 + (-4)(0.707 - j0.707) = -4 + 2.828 + j2.828$$

$$k = a - c(\omega_8^0) = 4 - 8(1) = -4$$

$$l = b - d(\omega_8^2) = -4 - (-4)(0.707 - j0.707) = -4 + 2.828 - j2.828$$

$$m = e + g(\omega_8^0)$$

$$i = a + c(\omega_8^0) = 4 + 8(1) = 12$$

$$j = b + d(\omega_8^2) = -4 + (-4)(-j) = -4 + 4j$$

$$k = a - c(\omega_8^0) = 4 - (8) = -4$$

$$l = b - d(\omega_8^2) = -4 - (-4)(-j) = -4 - 4j$$

$$m = e + g(\omega_8^0) = 6 + 10(1) = 16$$

$$n = f + h(\omega_8^2) = -4 + (-4)(-j) = -4 + 4j$$

$$o = e - g(\omega_8^0) = 6 - 10 = -4$$

$$p = f - h(\omega_8^2) = -4 - (-4)(-j) = -4 - 4j$$

$$x(0) = i + m(\omega_8^0) = 12 + 16(1) = 28$$

$$x(1) = J + n(\omega_8^1) = (-4 + 4j) + (-4 + 4j)(0.707 - j0.707) \\ = -4 + j9.656$$

$$x(2) = K + o(\omega_8^2) = -4 + (-4)(-j) = -4 + 4j$$

$$x(3) = L + p(\omega_8^3) = -4 - 4j + (-4 - 4j)(-0.707 - j0.707) = -4 + j9.656$$

$$x(4) = i - m(\omega_8^0) = 12 - 16 = -4$$

$$x(5) = J - n(\omega_8^1) = (-4 + 4j) - (-4 + 4j)(0.707 - j0.707) \\ = -4 - j9.656$$

$$x(6) = K - o(\omega_8^2) = -4 - (-4)(-j) = -4 - 4j$$

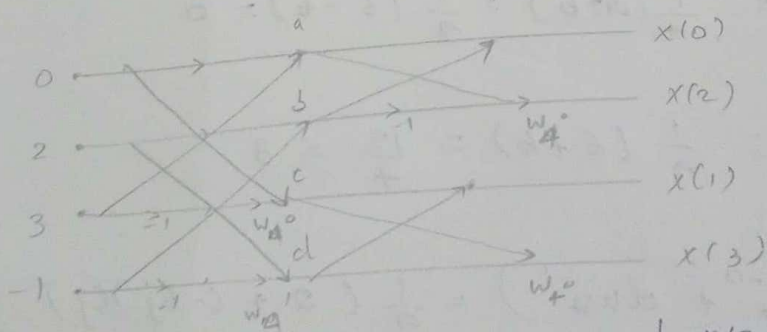
$$x(7) = L - p(\omega_8^3) = -4 - 4j - (-4 - 4j)(-0.707 - j0.707) \\ = -4 - j9.656$$

$$x(k) = \{28, -4 + j9.656, -4 + 4j, -4 + j9.656j, -4, -4 - j9.656, -4 - 4j, -4 - j9.656j\}$$

Determine N-pt DFT of the given sequence

$x(n) = \{0, 2, 3, -1\}$ using radix-2 DIF-FFT.

Soln
 $N=4$: $x(0)=0$, $x(1)=2$, $x(2)=3$, $x(3)=-1$



$$a = 0 + 3 = 3$$

$$b = (2 - 1)W_4^1 = 1e^{-j\pi/2} = -j$$

$$c = (0 - 3)W_4^0 = -3$$

$$d = (2 + 1)W_4^1 = 3(-j) = -3j$$

$$x(k) = \{4, -3 - 3j, 2, -3 + 3j\}$$

$$x(0) = a + b = 3 + j2 = 4$$

$$x(2) = (a - b)W_4^0 = 3 - 1 = 2$$

$$x(1) = c + d = -3 - 3j = -3 - 3j$$

$$x(3) = c - d = -3 + 3j = -3 + 3j$$

IDFT using DIF-FFT Algorithm

$$N\text{-pt IDFT } [X(K)] = x(n) = \frac{1}{N} \sum_{K=0}^{N-1} X(K) W_N^{-nK}$$

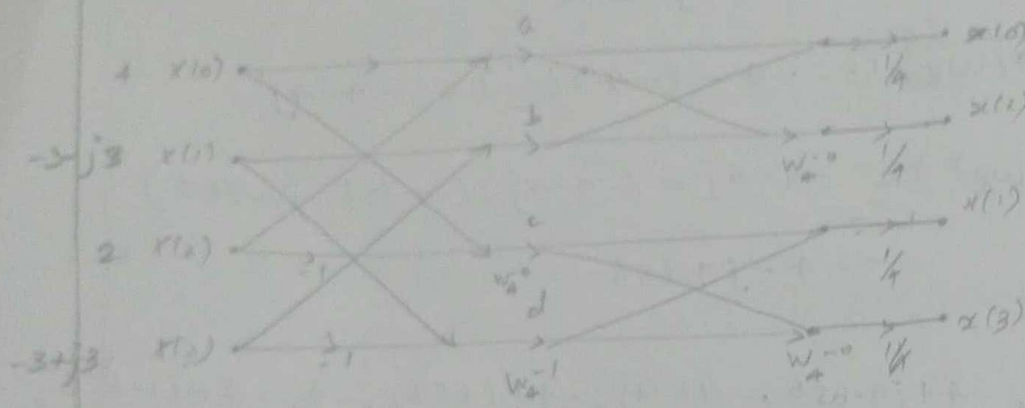
* $X(K) \rightarrow \text{ip}$

* consider complex conjugate of twiddle factor

* multiply the factor $\frac{1}{N}$ at each & every point at

o/p side

compute N-pt IDFT
let $X(K) = \{4, 0, -3-j3, 2, -3+j3\}$



$$W_4^0 = e^{j \frac{2\pi}{4} \cdot 0} = 1$$

$$W_4^{-1} = e^{-j \frac{2\pi}{4} \cdot 1} = e^{-j\pi/2} = -j$$

$$W_4^{-2} = e^{-j \frac{2\pi}{4} \cdot 2} = e^{-j\pi} = -1$$

$$W_4^{-3} = e^{-j \frac{2\pi}{4} \cdot 3} = e^{-j3\pi/2} = +j$$

$$a = 4 - 3 - j3 = 4 + 2 = 6$$

$$b = -3 - j3 - 3 + j3 = -6$$

$$c = 4 - 2 = 2$$

$$d = -3 - j3 + 3 - j3 = -6j$$

$$x(0) = \frac{a}{4} + \frac{b}{4} = \frac{1}{4} (a+b) = \frac{1}{4} (6-6) = 0$$

$$x(2) = \frac{(a-b)}{4} = \frac{1}{4} (6+6) = \frac{12}{4} = 3$$

$$x(1) = \frac{1}{4} (c W_4^{-0} + d W_4^{-1}) = \frac{1}{4} (2 + (-6j)(-j))$$

$$= \frac{1}{4} (2 + 6) = \frac{1}{4} \times 8 = 2$$

$$x(3) = \frac{1}{4} (c W_4^{-0} - d W_4^{-1}) = \frac{1}{4} (2 - (-6j)(-j)) = \frac{1}{4} (2 - 6) = \frac{1}{4} (-4) = -1$$

$$\therefore x(n) = \{0, 9, 3, -1\}$$